Supplementary Material for "FundusShift and LOTUS: Benchmarking and Improving Fundus Disease Classification at Unseen Sources"

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This supplement gives every table behind the main paper, plus analyses that did not fit there. All tables are generated from the saved results in the repository (`https://doi.org/10.5281/zenodo.23979471`) by `paper/latex/make_assets.py`, `make_supp.py` and `make_supp_extra.py`; LOTUS is the final method, and the *initial design* is LOTUS with the between-site projection of the main paper (Sec. 5); numbers are means over seeds unless stated. Abbreviations: bAcc, balanced accuracy; Eq-src, mean of per-source bAcc; R/T/F, raw, temperature scaling (TS), focal temperature scaling (FTS); LP, linear probe; LAC, least-ambiguous conformal sets; SGR, selective guaranteed risk; FR, disease images answered normal, over all disease images; G→N, glaucoma images accepted and predicted normal, over all glaucoma images.

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S1 Data

S1.1 Sources, label scope and licences

Table S1 lists every source with its class counts per split. "Scope" records which diseases a source graded: in DR-only sources a normal label means "no DR", and glaucoma was not assessed.

Table S1: Sources. Counts are N/D/G. Development sources contribute train, validation and calibration; test sources and Chákṣu contribute only the evaluation split.

Source	Country	Scope	Licence	Train	Validation	Calibration	Test or OOD
Bangladesh Eye	Bangladesh	full	CC BY 4.0	677/1200/925	85/150/116	84/150/115	0/0/0
OIA-DDR	China (147 hospitals, 23 pro	DR only	CC BY-NC-SA 4.0	5012/5004/0	627/626/0	626/625/0	0/0/0
DeepDRiD	China	DR only	CC BY-SA 4.0	732/868/0	94/110/0	88/108/0	0/0/0
GRAPE	China	full	CC0	0/0/470	0/0/58	0/0/58	0/0/0
IDRiD	India	DR only	CC BY 4.0	133/277/0	17/35/0	17/35/0	0/0/0
Jichi	Japan	DR only	CC BY 4.0	5028/2504/0	635/308/0	632/308/0	0/0/0
SUSTech-SYSU	China	DR only	CC BY 4.0	505/470/0	63/59/0	63/59/0	0/0/0
AFIO	Pakistan (Armed Forces In-sti	full	CC BY 4.0	–	–	–	67/18/44
FIVES	China	full	CC BY 4.0	–	–	–	193/200/193
HRF	Germany / Czech Republic	full	CC BY 4.0	–	–	–	15/15/15
HYGD	Israel	full	ODC-By 1.0	–	–	–	0/0/548
PALM	China (Zhongshan Oph-thalmic	full	CC0	–	–	–	161/0/0
PAPILA	Spain	full	GPL-3.0	–	–	–	333/0/87

Table S1: (continued)

Source	Country	Scope	Licence	Train	Validation	Calibration	Test or OOD
Paraguay	Paraguay	DR only	CC BY 4.0	–	–	–	708/723/0
Chákṣu	India	full	CC BY 4.0	–	–	–	1156/0/187

Reading. The development side holds 1,395 glaucoma training images, all from Bangladesh Eye (925) and GRAPE (470), whereas DR and normal images come from the six sources other than GRAPE. Five of seven development sources graded only DR. On the test side, only Paraguay is DR-only, HYGD is G-only and PALM is N-only, so per-source bAcc on those two reduces to one recall. These facts explain much of the glaucoma behaviour in the main paper.

S1.2 Source hierarchy, patients and label scope

FundusShift holds out whole public *collections* (sources), not hospitals. Table S2 gives the institution or country behind each collection, the number of patients where the source releases identifiers, the classes it contributes and what its "normal" label means. All but one collection come from a single institution or study; OIA-DDR pools 147 hospitals, so it is one development source with internal variety, not 147 held-out units. Each collection was also assigned an acquisition family (institution), and the audit of Sect. S1.5 confirms that no family spans the development and test sides; RFMiD 2.0, from the same institution as IDRiD, is used only for duplicate checks.

Table S2. Source hierarchy and label scope. Patients: distinct patient identifiers among used images (n/a: not released; for Chákṣu only the 60 Bosch-camera images carry identifiers, so its other images form leakage groups by duplicate detection alone and are treated as independent in bootstrap intervals). Normal means: what the source's normal label certifies.

Role	Source	Institution / country	Images	Patients	Classes	Normal means
Dev	Bangladesh Eye	Bangladesh	3,502	n/a	N/D/G	no graded disease
Dev	OIA-DDR	China (147 hospitals, 23 provinces)	12,520	n/a	N/D	no DR (DR-only grading)
Dev	DeepDRiD	China	2,000	500	N/D	no DR (DR-only grading)
Dev	GRAPE	China	586	133	G	no graded disease
Dev	IDRiD	India	514	n/a	N/D	no DR (DR-only grading)
Dev	Jichi	Japan	9,415	2,726	N/D	no DR (DR-only grading)
Dev	SUSTech-SYSU	China	1,219	n/a	N/D	no DR (DR-only grading)
Test	AFIO	Pakistan (Armed Forces Institute of Ophthalmology, Rawalpindi)	129	65	N/D/G	no graded disease
Test	FIVES	China	586	n/a	N/D/G	no graded disease
Test	HRF	Germany / Czech Republic	45	n/a	N/D/G	no graded disease
Test	HYGD	Israel	548	186	G	no graded disease
Test	PALM	China (Zhongshan Ophthalmic Center)	161	n/a	N	no graded disease
Test	PAPILA	Spain	420	210	N/G	no graded disease
Test	Paraguay	Paraguay	1,431	n/a	N/D	no DR (DR-only grading)
OOD	Chákṣu	India	1,343	30 (60 img.)	N/G	no graded disease

Reading. Five of seven development sources and one test source (Paraguay) graded DR only, so their normal images were never assessed for glaucoma. Only four test sources

assessed their normal images for glaucoma (AFIO, FIVES, HRF, which graded all three classes, and PAPILA, which graded N and G); Table S16 reports test bAcc restricted to them, and the label-scope experiment of Sect. S8 tests treating DR-only normals as possibly glaucomatous. Because glaucoma appears in only two development sources, glaucoma results measure transfer from those two sites, and the rotations (Table S29) show how they change with more glaucoma sources.

S1.3 Image quality by source

Table S3. Image properties of used images by source (medians): original size, field-of-view fraction, brightness (mean luminance in the field), RMS contrast, Laplacian sharpness, and the share flagged as very dark, overexposed, low field of view or blurry for its source.

Source	Size	FOV	Brightness	Contrast	Sharpness	Flagged
Bangladesh Eye	2,004×1,690	.838	111.2	22.5	65.0	0.8%
OIA-DDR	2,592×1,824	.535	83.4	20.8	61.6	0.8%
DeepDRiD	1,736×1,824	.682	86.2	29.4	38.2	2.1%
GRAPE	2,136×2,136	.756	62.6	20.0	31.4	1.4%
IDRiD	4,288×2,848	.692	87.0	17.8	47.1	1.2%
Jichi	848×848	.788	99.4	17.1	84.2	1.2%
SUSTech-SYSU	2,880×2,136	.896	109.9	13.7	54.7	1.1%
AFIO	1,612×1,536	.651	55.7	15.8	70.8	3.1%
FIVES	2,048×2,048	.758	53.2	17.0	52.2	3.2%
HRF	3,504×2,336	.845	92.7	17.4	44.1	2.2%
HYGD	1,897×1,897	.787	62.2	20.5	11.0	1.5%
PALM	2,124×2,056	.804	55.4	17.0	28.3	0.0%
PAPILA	2,576×1,934	.809	61.7	13.3	26.1	1.9%
Paraguay	2,124×2,056	.805	39.0	10.2	16.2	1.0%
Chákṣu	2,448×3,264	.465	55.6	24.6	211.3	1.0%

Reading. Sources differ strongly in resolution, field of view, exposure and sharpness, and these trivial statistics alone identify the source; flagged images are rare (0–3%). The hardest test sources are not simply the lowest-quality ones: AFIO and FIVES are dark, but Paraguay, the darkest and among the least sharp, is the easiest for every method. Source difficulty therefore reflects class composition and label scope as well as appearance, and we do not attribute it to image quality alone.

S1.4 Mapping from source labels

Table S4: Native labels mapped to each harmonized class (first token of the stored label; device and vote suffixes omitted).

Source	Class	Native labels
Bangladesh Eye	N	Healthy
Bangladesh Eye	D	Diabetic_Retinopathy
Bangladesh Eye	G	Glaucoma
OIA-DDR	N	ddr_0
OIA-DDR	D	ddr_1, ddr_2, ddr_3, ddr_4

Table S4: (continued)

Source	Class	Native labels
DeepDRiD	N	icdr_0
DeepDRiD	D	icdr_1, icdr_2, icdr_3, icdr_4
GRAPE	G	glaucoma_follow_up
IDRiD	N	icdr_0
IDRiD	D	icdr_1, icdr_2, icdr_3, icdr_4
Jichi	N	davis_ndr
Jichi	D	davis_pdr, davis_ppdr, davis_sdr
SUSTech-SYSU	N	icdr_0
SUSTech-SYSU	D	icdr_1, icdr_2, icdr_3, icdr_4, laser_scars(reclassified_icdr_2), laser_scars(reclassified_icdr_3), ...
AFIO	N	macula_healthy, od_healthy
AFIO	D	macula_ci-dme
AFIO	G	od_glaucoma
FIVES	N	normal
FIVES	D	diabetic_retinopathy
FIVES	G	glaucoma
HRF	N	healthy
HRF	D	diabetic_retinopathy
HRF	G	glaucoma
HYGD	G	GON+
PALM	N	normal
PAPILA	N	healthy
PAPILA	G	glaucoma
Paraguay	N	1._No_DR_signs
Paraguay	D	2._Mild_(or_early)_NPDR, 3._Moderate_NPDR, 4._Severe_NPDR, 5._Very_Severe_NPDR, 6._PDR, 7._Advanced_PDR
Chákṣu	N	normal
Chákṣu	G	glaucoma_suspect

Reading. A native label is mapped only when it supports the class. Grading scales differ: ICDR grades 1–4, Davis SDR–PDR and named DR categories all become D. Glaucoma suspects in Chákṣu follow the source's majority vote. HYGD's 199 GON− images were excluded rather than mapped to N, because "no glaucomatous optic neuropathy" does not certify the absence of other disease; this makes HYGD single-class. The glaucoma reference standard differs between sources: clinical work-up in GRAPE (visual-field follow-up patients) and HYGD (GON+), image-based expert grading in Chákṣu (majority vote, including suspects), and not reported for Bangladesh Eye and AFIO, so G pools definitions of different strength. D pools all DR grades from mild NPDR upward; referable DR is not separated (recall by grade: Table S42). The 861-image open-set probe holds images of other conditions from PALM (637), FIVES (200) and AFIO (24), and is distinct from the 860 RFMiD 2.0 images excluded from the core task.

S1.5 Leakage control and audit

Duplicates were detected by exact SHA-256 hashes and by a vessel-pattern signature (band-pass green channel, 32×32) with shift-tolerant normalized cross-correlation at threshold 0.70, then joined with patient identifiers into leakage groups. An automated audit of the released splits found zero violations for each of: a leakage group in more than one split; an identical file in more than one split; a patient in more than one split; a near-duplicate pair across splits; an acquisition family on both development and test sides; a source on both sides; a calibration image without a Cal-A/Cal-B assignment.

Brightness-altered copies of GRAPE photographs inside the FIVES glaucoma class were found this way and removed. The detector linked 4,247 image pairs (678 byte-identical, 29 across sources), forming 2,450 leakage groups with more than one image. Of 40,414 inventoried images, 35,280 enter the protocol; the rest were removed for an excluded or unmapped native label (1,820 and 1,453), duplicates whose copies carried conflicting labels (956), a source excluded from the core task (860), or duplication of a test-side image (45). The code is `fsbench/dedup.py` and `fsbench/vessel.py`.

Table S5. Duplicate links found by the detector (exact hash or vessel-pattern NCC ≥ 0.70), by source pair: links, byte-identical links, and links whose two images carry different labels.

Scope	Source(s)	Links	Exact	Label conflict
across	FIVES – GRAPE	29	0	0
within	Jichi	2,691	0	316
within	Bangladesh Eye	770	484	612
within	OIA-DDR	203	106	1
within	GRAPE	189	0	0
within	Paraguay	181	19	3
within	SUSTech-SYSU	55	0	0
within	DeepDRiD	38	0	0
within	PALM	22	21	6
within	FIVES	18	6	7
within	AFIO	16	16	12
within	HYGD	11	10	2
within	RFMiD 2.0	10	3	4
within	Chákṣu	8	7	1
within	IDRiD	6	6	1

Reading. All 29 cross-source links join GRAPE photographs to brightness-altered copies inside FIVES, a development–test leak that hashes alone would miss. Within-source links mostly join repeated visits or re-exports; links with conflicting labels concentrate in Bangladesh Eye and Jichi, and the 956 images removed for conflicting labels (above) come from such groups.

Excluded collections. RFMiD 2.0 was used for inventory and duplicate checks only, because class is confounded with camera. A glaucoma collection of optic-disc crops was excluded for its different field of view. OIA-DDR is CC BY-NC-SA 4.0, so derived data remain non-commercial.

Ethics. All images are publicly released; this is a secondary analysis. Ethics and consent statements were found in the publications of FIVES, GRAPE, PAPILA, SUSTech-SYSU, DeepDRiD, Jichi, HYGD, Chákṣu, PALM and Paraguay. Statements were not found on the release pages of HRF, Bangladesh Eye and AFIO. Age and sex are available only for PAPILA, so subgroup fairness cannot be assessed.

S2 Models, Recipes and Prompts

Table S6. Encoders. "Fine-tuned" marks the 13 architectures trained end to end; every encoder was also probed frozen.

Encoder	Pretraining data	Fine-tuned
deit_b16	ImageNet-1k	yes
convnext_t	ImageNet-1k	yes
swin_t	ImageNet-1k	yes
inception_v3	ImageNet-1k	yes
efficientnet_b0	ImageNet-1k	yes
densenet121	ImageNet-1k	yes
vit_b16_in21k	ImageNet-21k	yes
resnet50	ImageNet-1k	yes
mobilenetv3_l	ImageNet-1k	yes
vgg16	ImageNet-1k	yes
vit_b16_mae	ImageNet-1k (MAE, self-supervised)	yes
dinov2_b14	LVD-142M (self-supervised)	no
retfound_mae	MEH-MIDAS + public fundus (MAE, RETFound)	yes
retfound_dinov2	MEH-MIDAS fundus (DINOv2, RETFound)	yes
clip_b16	OpenAI web image-text	no
metaclip_b16	MetaCLIP (web image-text)	no
siglip_b16	WebLI	no
siglip_l16	WebLI	no
siglip2_b32	WebLI	no
siglip2_b16	WebLI	no
siglip2_l16	WebLI	no
bioclip_b16	iNat/biodiversity (TreeOfLife-10M)	no
bioclip2_l14	biodiversity (TreeOfLife-200M)	no
biomedclip	PMC-15M biomedical figures	no
medsiglip	Medical image-text (MedGemma report); fundus: EyePACS	no
flair	Retinal image-text; includes five FundusShift test sources	no

Table S7. Training recipes. Every setting selects by validation bAcc unless stated; post-hoc calibration always uses Cal-A, thresholds Cal-B.

Fine-tuning	Native input size and normalization; AdamW, lr 10^{-4} (head 10^{-3}), wd 0.05, cosine with one warm-up epoch, 15 epochs, batch 32; (source, class)-balanced sampling; crop, flip, rotation and colour augmentation; mixed precision; five losses (CE; label smoothing 0.1; focal γ=3; MDCA β=1; MbLS m=10, λ=0.1); 3 seeds. NLL-rule arm: checkpoint by validation NLL.
Linear, MLP probes	Frozen embeddings standardized on train; AdamW lr 10^{-3}, wd 10^{-4}, 40 epochs, batch 256; weighting none, class or (class, source); MLP: one hidden layer of 512.
16-shot probe	16 images per class, one per leakage group, redrawn per seed, 300 epochs.
Zero-shot	Three prompt banks (Table S8); bank chosen by validation bAcc.
CoOp	16 learnable context tokens, frozen towers, SGD lr 0.002, cosine, 200 full-batch steps on the 16 shots; ctx-init: 4 tokens from "a photo of a"; CoOp+LP: then a linear head, 200 steps, lr 10^{-3}; loss variants CE, focal, LS, MbLS, MDCA.
Tip-Adapter	Official recipe on the same shots; (α, β) grid and bank by validation bAcc (official accuracy rule kept as sensitivity); -F: keys fine-tuned 20 epochs, AdamW lr 10^{-3}.
ERM, Group DRO, CORAL	Fine-tuning recipe with ConvNeXt-T, or DINOv2-B/14 with LoRA rank 8 on query and value; Group DRO η=0.01, group = development source; CORAL λ=1.
SWAD	Dense averaging over segments of 150 updates, official loss-valley rule (3, 6, 0.3); replaces the validation checkpoint.
WiSE-FT	Head from the zero-shot classifier, AdamW lr 10^{-5}, wd 0.1, 15 epochs; interpolation $\alpha \in \{0, 0.1, \ldots, 1\}$ chosen by validation bAcc.
Multiple choice	Image plus a three-option question; next-token log-likelihoods of the option letters averaged over all six option orders.
LOTUS	Table S44.

Table S8: Prompt banks used by every zero-shot model and by the text branch of LOTUS.

Bank	Prompts
CLIP ensemble	a photo of .; a colour fundus photograph of .; a retinal image of .; a fundus photo showing .; an ophthalmology image of .; a close-up retinal photograph of .; . with class names: N: a normal retina, D: diabetic retinopathy, G: glaucoma
CuPL (N)	a healthy fundus with a pink optic disc, a clear macula and regular retinal vessels. a normal retina without haemorrhages, exudates or optic disc abnormality. a fundus photograph with an evenly coloured background and a well-defined foveal reflex. a retina with a small physiological optic cup and an intact neural rim. a fundus free of microaneurysms, cotton-wool spots and neovascularisation.
CuPL (D)	a retina with scattered microaneurysms and dot-blot haemorrhages. a fundus with yellow hard exudates near the macula. a retinal photograph with cotton-wool spots and venous beading. a fundus with abnormal new vessels and preretinal haemorrhage. a diabetic eye with intraretinal haemorrhages in several quadrants.
CuPL (G)	an optic disc with a large cup-to-disc ratio. a fundus with a thinned or notched neural rim of the optic disc. a retina with retinal nerve fibre layer defects next to the optic disc. an optic nerve head with vertical cupping and pallor. a fundus with peripapillary atrophy and glaucomatous optic neuropathy.
Single (N)	a colour fundus photograph of a healthy retina with a normal optic disc and no lesions.
Single (D)	a colour fundus photograph with diabetic retinopathy showing microaneurysms, haemorrhages and exudates.
Single (G)	a colour fundus photograph with glaucoma showing an enlarged optic cup and a thinned neural rim.

S3 Complete Benchmark Results

Table S9: Every setting and encoder, with the configuration of highest validation bAcc. †FLAIR's pretraining includes five FundusShift test sources. Multiple-choice models have no validation scores and no probabilities.

Setting	Encoder	Config	Val	Test	Eq-src	Chákṣu	ECE R/T/F	NLL R/T/F
16-shot probe	flair†	none/mbls	.809	.750	.684	.683	.134/.128/.127	.947/.915/.891
16-shot probe	medsiglip	none/ce	.802	.748	.683	.675	.096/.044/.037	.701/.644/.644
16-shot probe	siglip2_l16	none/mbls	.696	.621	.564	.195	.124/.075/.074	.999/.885/.889
16-shot probe	biomedclip	none/ce	.677	.479	.430	.450	.198/.113/.120	1.18/1.05/1.06
16-shot probe	dinov2_b14	none/ce	.675	.529	.503	.192	.275/.080/.088	1.53/.993/.996
16-shot probe	efficientnet_b0	none/ce	.665	.482	.458	.279	.216/.081/.085	1.24/1.02/1.03
16-shot probe	retfound_mae	none/mbls	.662	.373	.339	.207	.437/.238/.241	1.94/1.17/1.17
16-shot probe	siglip_l16	none/ce	.658	.440	.414	.316	.314/.166/.168	1.51/1.16/1.16
16-shot probe	siglip_b16	none/ce	.653	.526	.516	.266	.175/.093/.102	1.24/1.02/1.03
16-shot probe	densenet121	none/ce	.653	.514	.499	.347	.207/.063/.056	1.15/.963/.955
16-shot probe	convnext_t	none/ce	.649	.523	.520	.403	.214/.063/.069	1.24/.971/.970
16-shot probe	clip_b16	none/mbls	.649	.486	.419	.251	.192/.103/.108	1.24/1.08/1.09
16-shot probe	deit_b16	none/mbls	.645	.517	.484	.233	.228/.096/.097	1.24/1.01/1.01
16-shot probe	retfound_dinov2	none/ce	.644	.437	.426	.158	.282/.087/.095	1.47/1.09/1.09
16-shot probe	mobilenetv3_l	none/ce	.643	.450	.426	.357	.277/.106/.113	1.48/1.08/1.09
16-shot probe	siglip2_b16	none/ce	.638	.456	.422	.281	.315/.137/.141	1.85/1.11/1.11
16-shot probe	swin_t	none/ce	.634	.494	.458	.204	.214/.059/.062	1.21/1.00/1.01
16-shot probe	vit_b16_in21k	none/ce	.630	.495	.469	.268	.234/.087/.090	1.34/1.02/1.02
16-shot probe	resnet50	none/mbls	.628	.541	.509	.429	.149/.070/.061	1.12/.986/.979
16-shot probe	inception_v3	none/mbls	.627	.457	.425	.437	.323/.099/.101	1.59/1.07/1.08
16-shot probe	siglip2_b32	none/ce	.622	.471	.439	.375	.260/.098/.098	1.41/1.07/1.07
16-shot probe	vit_b16_mae	none/ls	.621	.392	.412	.239	.331/.189/.200	1.57/1.20/1.20
16-shot probe	vgg16	none/ls	.619	.468	.459	.378	.213/.135/.125	1.73/1.24/1.19
16-shot probe	metaclip_b16	none/ce	.613	.612	.539	.458	.145/.085/.079	1.10/.906/.903
16-shot probe	bioclip2_l14	none/ce	.612	.495	.467	.407	.247/.064/.068	1.48/1.02/1.03
16-shot probe	bioclip_b16	none/ce	.597	.411	.386	.204	.347/.142/.141	1.64/1.14/1.14
CoOp	siglip_b16	ce	.660	.491	.458	.318	.236/.175/.177	1.35/1.18/1.16
CoOp	biomedclip	ls	.654	.475	.437	.379	.235/.163/.173	1.37/1.19/1.21
CoOp	clip_b16	ls	.631	.448	.382	.206	.235/.189/.190	1.27/1.16/1.17
CoOp	bioclip2_l14	mbls	.622	.474	.462	.270	.235/.133/.134	1.50/1.21/1.24
CoOp	siglip_l16	ce	.612	.403	.479	.368	.107/.189/.191	1.13/1.25/1.28
CoOp	bioclip_b16	mdca	.593	.415	.422	.152	.429/.154/.155	2.59/1.19/1.20
CoOp	metaclip_b16	ls	.585	.540	.483	.358	.125/.089/.084	.990/.951/.944
CoOp	siglip2_l16	zs_penalty	.408	.459	.489	.444	.099/.044/.045	1.01/.994/.997
CoOp	siglip2_b32	zs_penalty	.401	.399	.315	.427	.045/.191/.186	1.05/1.11/1.10
CoOp	siglip2_b16	zs_penalty	.358	.335	.233	.205	.132/.224/.224	1.10/1.16/1.16
CoOp ctx-init	biomedclip	mdca	.642	.497	.447	.384	.287/.150/.160	1.76/1.19/1.22
CoOp ctx-init	clip_b16	mdca	.638	.499	.429	.171	.189/.133/.134	1.20/1.08/1.08
CoOp ctx-init	metaclip_b16	mdca	.634	.605	.534	.418	.092/.043/.042	.884/.838/.836
CoOp ctx-init	bioclip2_l14	mdca	.611	.468	.458	.231	.188/.115/.115	1.36/1.18/1.20
CoOp ctx-init	bioclip_b16	mbls	.583	.444	.418	.157	.419/.173/.175	2.56/1.21/1.22
CoOp ctx-init	siglip_l16	zs_penalty	.442	.421	.519	.480	.075/.092/.097	1.08/1.09/1.09
CoOp ctx-init	siglip_b16	mdca	.421	.458	.513	.492	.048/.120/.119	1.09/1.06/1.06
CoOp ctx-init	siglip2_l16	mdca	.386	.361	.450	.499	.231/.183/.178	1.29/1.20/1.23
CoOp ctx-init	siglip2_b32	zs_penalty	.358	.380	.447	.518	.054/.061/.058	1.06/1.06/1.06

Table S9: (continued)

Setting	Encoder	Config	Val	Test	Eq-src	Chákṣu	ECE R/T/F	NLL R/T/F
CoOp ctx-init	siglip2_b16	focal	.351	.341	.424	.496	.178/.157/.159	1.24/1.20/1.22
CoOp + LP	siglip_b16	mbls	.668	.496	.463	.314	.243/.176/.179	1.37/1.18/1.17
CoOp + LP	biomedclip	mdca	.652	.482	.433	.368	.292/.177/.186	1.68/1.22/1.24
CoOp + LP	siglip_l16	ce	.647	.447	.490	.327	.101/.169/.172	1.12/1.23/1.27
CoOp + LP	clip_b16	ls	.637	.449	.384	.213	.244/.190/.191	1.29/1.16/1.16
CoOp + LP	siglip2_l16	ls	.632	.559	.543	.276	.154/.030/.030	.969/.897/.895
CoOp + LP	metaclip_b16	mbls	.628	.594	.506	.410	.170/.067/.066	.959/.860/.857
CoOp + LP	bioclip2_l14	mbls	.623	.477	.463	.276	.253/.136/.136	1.59/1.21/1.23
CoOp + LP	bioclip_b16	mdca	.595	.415	.424	.156	.433/.156/.157	2.63/1.19/1.20
CoOp + LP	siglip2_b32	ce	.539	.487	.432	.459	.068/.064/.061	1.00/.998/.995
CoOp + LP	siglip2_b16	mbls	.510	.419	.369	.417	.051/.075/.073	1.04/1.04/1.04
CORAL	convnext_t	ce	.883	.676	.623	.349	.198/.136/.148	1.38/1.01/1.11
CORAL	dinov2_lora	ce	.853	.667	.625	.340	.143/.144/.150	1.05/1.05/1.11
ERM	dinov2_lora	ce	.857	.707	.646	.419	.154/.115/.119	1.12/.944/.965
Fine-tune	swin_t	mbls	.893	.655	.600	.298	.223/.171/.182	1.56/1.17/1.31
Fine-tune	convnext_t	ls	.888	.676	.611	.319	.144/.168/.177	.926/.982/1.04
Fine-tune	retfound_mae	mdca	.887	.525	.417	.291	.370/.276/.288	2.69/1.50/1.57
Fine-tune	retfound_dinov2	mbls	.886	.583	.501	.358	.285/.192/.207	1.93/1.19/1.33
Fine-tune	densenet121	ce	.882	.626	.581	.309	.238/.139/.158	1.88/1.13/1.30
Fine-tune	resnet50	mdca	.881	.666	.620	.319	.184/.110/.126	1.48/1.05/1.17
Fine-tune	inception_v3	mbls	.877	.664	.624	.405	.200/.139/.147	1.25/.956/1.05
Fine-tune	vgg16	mbls	.875	.671	.625	.371	.169/.135/.147	1.28/1.10/1.24
Fine-tune	deit_b16	mdca	.868	.622	.560	.313	.230/.158/.167	1.88/1.28/1.42
Fine-tune	vit_b16_in21k	mdca	.863	.596	.535	.336	.251/.180/.190	2.02/1.31/1.42
Fine-tune	efficientnet_b0	ce	.861	.651	.596	.266	.188/.108/.122	1.34/.974/1.04
Fine-tune	vit_b16_mae	ce	.856	.587	.532	.302	.272/.196/.205	2.17/1.42/1.59
Fine-tune	mobilenetv3_l	ls	.846	.648	.596	.238	.082/.130/.144	.855/.923/.993
Fine-tune, NLL rule	convnext_t	ce	.870	.677	.619	.324	.185/.133/.147	1.35/1.04/1.16
Fine-tune, NLL rule	resnet50	ce	.857	.647	.601	.254	.192/.117/.133	1.57/1.12/1.28
Group DRO	convnext_t	ce	.849	.707	.649	.417	.082/.091/.101	.886/.926/1.04
Group DRO	dinov2_lora	ce	.798	.733	.677	.486	.050/.035/.034	.732/.707/.732
Linear probe	medsiglip	class/ls	.920	.728	.654	.347	.024/.079/.093	.647/.694/.739
Linear probe	flair†	class/focal	.900	.721	.693	.703	.088/.117/.117	.705/.777/.775
Linear probe	dinov2_b14	class/mbls	.889	.674	.616	.434	.104/.103/.115	.843/.842/.875
Linear probe	retfound_mae	class/ls	.886	.455	.351	.243	.368/.412/.414	1.43/1.71/1.69
Linear probe	retfound_dinov2	class/ce	.868	.606	.522	.115	.185/.147/.158	1.16/1.03/1.08
Linear probe	siglip2_l16	class/mbls	.866	.605	.562	.206	.167/.159/.170	1.11/1.08/1.16
Linear probe	vit_b16_mae	class/ls	.866	.579	.536	.102	.136/.190/.200	.924/1.02/1.03
Linear probe	vit_b16_in21k	class/ls	.861	.608	.572	.292	.119/.170/.182	.983/1.09/1.15
Linear probe	metaclip_b16	class/ls	.861	.599	.551	.101	.114/.177/.188	.908/1.01/1.05
Linear probe	siglip2_b16	class/ls	.860	.546	.520	.484	.130/.187/.201	1.04/1.15/1.22
Linear probe	siglip_b16	class/ls	.856	.567	.540	.169	.130/.189/.201	.988/1.10/1.13
Linear probe	bioclip2_l14	class/ls	.853	.538	.520	.365	.164/.212/.227	1.05/1.15/1.17
Linear probe	densenet121	class/mdca	.852	.569	.522	.121	.228/.203/.211	1.38/1.26/1.35
Linear probe	siglip_l16	class/ce	.851	.524	.508	.352	.237/.235/.246	1.46/1.45/1.57
Linear probe	swin_t	class/mbls	.850	.584	.523	.176	.198/.180/.191	1.20/1.14/1.17
Linear probe	convnext_t	class/mbls	.848	.615	.553	.218	.145/.137/.147	.999/.981/1.01
Linear probe	clip_b16	class/ls	.844	.595	.528	.477	.104/.162/.171	.950/1.07/1.14
Linear probe	deit_b16	class/mdca	.844	.595	.537	.272	.179/.155/.167	1.11/1.05/1.09
Linear probe	bioclip_b16	class/mbls	.830	.539	.479	.164	.207/.198/.212	1.22/1.19/1.23
Linear probe	mobilenetv3_l	class/ls	.825	.583	.525	.137	.089/.153/.165	.968/1.07/1.11
Linear probe	siglip2_b32	class/ls	.824	.554	.517	.461	.130/.177/.188	1.01/1.10/1.13
Linear probe	efficientnet_b0	class/ce	.824	.598	.534	.196	.210/.179/.185	1.32/1.19/1.18
Linear probe	resnet50	class/mbls	.815	.650	.576	.334	.081/.056/.060	.888/.858/.868
Linear probe	biomedclip	class/mdca	.803	.479	.459	.510	.274/.271/.281	1.34/1.33/1.36
Linear probe	inception_v3	class/mbls	.799	.543	.500	.347	.220/.190/.198	1.22/1.14/1.15
Linear probe	vgg16	class/mbls	.793	.534	.480	.267	.234/.157/.179	1.89/1.56/1.36
MLP probe	dinov2_b14	none/ce	.870	.598	.524	.318	.263/.172/.183	2.31/1.26/1.42
MLP probe	siglip2_l16	none/ce	.859	.559	.481	.140	.281/.194/.194	2.92/1.73/1.73
MLP probe	metaclip_b16	none/ce	.853	.524	.473	.122	.360/.260/.269	3.04/1.53/1.69
MLP probe	siglip_b16	none/ce	.851	.570	.542	.330	.279/.182/.189	2.71/1.42/1.57
MLP probe	siglip_l16	none/ce	.847	.546	.469	.355	.346/.212/.216	4.72/1.86/2.02
MLP probe	siglip2_b16	none/ce	.845	.542	.510	.431	.292/.205/.217	2.95/1.71/2.05
MLP probe	bioclip2_l14	none/ce	.838	.514	.457	.318	.349/.234/.245	3.01/1.53/1.66
MLP probe	clip_b16	none/ce	.837	.545	.487	.450	.344/.250/.256	4.01/1.94/2.18
MLP probe	bioclip_b16	none/ce	.827	.502	.434	.089	.328/.252/.255	2.14/1.50/1.56
MLP probe	siglip2_b32	none/ce	.817	.546	.503	.490	.302/.181/.189	3.26/1.52/1.74
MLP probe	biomedclip	none/ce	.806	.525	.506	.475	.272/.185/.194	2.84/1.85/2.08
SWAD	convnext_t	ce	.883	.672	.622	.309	.208/.139/.149	1.57/1.06/1.16
SWAD	dinov2_lora	ce	.832	.716	.655	.436	.127/.092/.097	.964/.854/.879
Tip-Adapter	biomedclip	–	.597	.478	.491	.617	.332/.091/.096	1.75/1.02/1.03
Tip-Adapter	siglip2_l16	–	.550	.497	.534	.533	.140/.128/.117	1.00/.930/.914
Tip-Adapter	clip_b16	–	.473	.456	.466	.424	.082/.071/.077	1.04/1.08/1.07
Tip-Adapter (acc. rule)	biomedclip	–	.597	.478	.491	.617	.332/.091/.096	1.75/1.02/1.03
Tip-Adapter (acc. rule)	siglip2_l16	–	.550	.497	.534	.533	.140/.128/.117	1.00/.930/.914
Tip-Adapter (acc. rule)	clip_b16	–	.473	.456	.466	.424	.082/.071/.077	1.04/1.08/1.07
Tip-Adapter-F	biomedclip	–	.666	.495	.453	.532	.225/.147/.152	1.28/1.10/1.11
Tip-Adapter-F	siglip2_l16	–	.602	.538	.551	.528	.086/.157/.155	1.09/.986/.969
Tip-Adapter-F	clip_b16	–	.564	.508	.511	.459	.109/.095/.090	1.05/.997/.995
Tip-Adapter-F (acc. rule)	biomedclip	–	.666	.495	.453	.532	.225/.147/.152	1.28/1.10/1.11
Tip-Adapter-F (acc. rule)	siglip2_l16	–	.602	.538	.551	.528	.086/.157/.155	1.09/.986/.969
Tip-Adapter-F (acc. rule)	clip_b16	–	.564	.508	.511	.459	.109/.095/.090	1.05/.997/.995
WiSE-FT	siglip2_l16	fully-tuned	.886	.634	.587	.406	.236/.168/.179	1.84/1.24/1.38
Zero-shot	flair†	descriptive_single	.653	.712	.624	.567	.162/.154/.137	1.09/.867/.827
Zero-shot	medsiglip	descriptive_single	.650	.709	.606	.615	.207/.032/.032	.838/.690/.690
Zero-shot	biomedclip	descriptive_single	.503	.471	.535	.562	.229/.061/.074	1.29/1.00/1.02
Zero-shot	metaclip_b16	descriptive_single	.476	.403	.406	.374	.093/.119/.119	1.06/1.07/1.07

Table S9: (continued)

Setting	Encoder	Config	Val	Test	Eq-src	Chákṣu	ECE R/T/F	NLL R/T/F
Zero-shot	siglip2_b32	descriptive_single	.473	.463	.501	.505	.028/.069/.063	1.05/1.07/1.07
Zero-shot	siglip2_l16	clip_ensemble	.461	.481	.494	.374	.031/.070/.085	1.06/1.17/1.13
Zero-shot	siglip_b16	cupl	.437	.433	.508	.503	.068/.090/.092	.959/.975/.973
Zero-shot	siglip2_b16	descriptive_single	.418	.509	.539	.518	.107/.176/.167	1.06/1.07/1.06
Zero-shot	bioclip_b16	descriptive_single	.366	.457	.518	.525	.358/.182/.183	3.79/1.10/1.10
Zero-shot	siglip_l16	cupl	.364	.357	.453	.500	.280/.162/.165	1.24/1.09/1.10
Zero-shot	bioclip2_l14	clip_ensemble	.344	.362	.281	.074	.326/.180/.186	1.54/1.16/1.18
Zero-shot	clip_b16	clip_ensemble	.327	.394	.445	.173	.055/.096/.094	1.10/1.15/1.16
Multiple choice	Fundus-R1-7B	avg6	–	.576	.569	.463	–	–
Multiple choice	Qwen2.5-VL-7B-Instruct	avg6	–	.424	.462	.488	–	–
Multiple choice	llava-med-v1.5-mistral-7b-hf	avg6	–	.347	.448	.507	–	–
Multiple choice	llava-onevision-qwen2-7b-ov-hf	avg6	–	.407	.327	.096	–	–

Reading. Within each setting the encoder ranking on test differs from that on validation (Sect. S5). MedSigLIP is the strongest frozen encoder across probe settings, and FLAIR is comparable only on sources it did not see in pretraining. Fine-tuned RETFound reaches high validation (0.886–0.887) but 0.525–0.583 on test, despite fundus pretraining. TS reduces ECE and NLL for most fine-tuned models but leaves ECE far above that of frozen probes. For transparency, the best frozen probe judged by test itself, a class-weighted cross-entropy MedSigLIP linear probe, reached test bAcc 0.778, equal-source 0.702, raw ECE 0.065 and recall G 0.657; validation selected the label-smoothed variant instead.

S4 Per-Source and Per-Class Results

Table S10: bAcc per unseen source (raw), equal-source mean and worst test source, for LOTUS and the validation-selected representatives.

Method	Paraguay	HYGD	FIVES	PAPILA	PALM	HRF	AFIO	Chákṣu	Eq-src	Worst
LOTUS	.919	.822	.721	.664	.836	.622	.652	.695	.748	.622
16-shot probe, medsiglip	.874	.756	.668	.499	.723	.711	.548	.675	.683	.499
CoOp, siglip_b16	.728	.130	.489	.197	.768	.415	.476	.318	.458	.130
CoOp ctx-init, biomedclip	.753	.038	.481	.435	.563	.400	.456	.384	.447	.038
CoOp + LP, siglip_b16	.735	.131	.496	.197	.781	.430	.471	.314	.463	.131
CORAL, convnext_t	.929	.263	.688	.413	.911	.622	.533	.349	.623	.263
ERM, dinov2_lora	.884	.505	.596	.498	.816	.607	.614	.419	.646	.498
Fine-tune, swin_t	.913	.237	.647	.468	.836	.600	.501	.298	.600	.237
Fine-tune, NLL rule, convnext_t	.917	.283	.704	.449	.880	.585	.516	.324	.619	.283
Group DRO, convnext_t	.920	.316	.704	.592	.870	.600	.539	.417	.649	.316
Linear probe, medsiglip	.934	.477	.559	.493	.907	.637	.571	.347	.654	.477
MLP probe, dinov2_b14	.889	.092	.631	.285	.747	.533	.492	.318	.524	.092
SWAD, convnext_t	.928	.243	.709	.409	.901	.615	.550	.309	.622	.243
Tip-Adapter, biomedclip	.399	.744	.427	.496	.480	.400	.493	.617	.491	.399
Tip-Adapter-F, biomedclip	.624	.283	.465	.471	.472	.407	.448	.532	.453	.283
WiSE-FT, siglip2_l16	.881	.250	.626	.423	.801	.600	.529	.406	.587	.250
Zero-shot, medsiglip	.729	.719	.723	.622	.379	.556	.512	.615	.606	.379

Table S11: Class recalls on the multi-class test sources.

	FIVES			AFIO			PAPILA		Paraguay	
Method	N	D	G	N	D	G	N	G	N	D
LOTUS	.606	.767	.789	.448	.833	.674	.899	.429	.893	.946

Table S11: (continued)

Method	FIVES			AFIO			PAPILA		Paraguay	
	N	D	G	N	D	G	N	G	N	D
16-shot probe, medsiglip	.418	.757	.831	.075	.796	.773	.139	.858	.797	.951
CoOp, siglip_b16	.477	.717	.275	.164	1.00	.265	.110	.284	.624	.833
CoOp ctx-init, biomedclip	.522	.840	.083	.373	.926	.068	.349	.521	.653	.852
CoOp + LP, siglip_b16	.530	.723	.233	.144	.981	.288	.121	.272	.625	.845
CORAL, convnext_t	.440	.700	.924	.687	.685	.227	.405	.421	.920	.937
ERM, dinov2_lora	.171	.683	.933	.443	.815	.583	.413	.582	.806	.963
Fine-tune, swin_t	.328	.702	.910	.483	.907	.114	.318	.617	.872	.954
Fine-tune, NLL rule, convnext_t	.525	.710	.877	.597	.722	.227	.395	.502	.896	.937
Group DRO, convnext_t	.560	.695	.858	.458	.833	.326	.748	.437	.917	.922
Linear probe, medsiglip	.169	.692	.817	.647	.944	.121	.891	.096	.917	.951
MLP probe, dinov2_b14	.380	.650	.864	.701	.722	.053	.451	.119	.879	.898
SWAD, convnext_t	.520	.715	.893	.657	.759	.235	.362	.456	.916	.940
Tip-Adapter, biomedclip	.245	.362	.675	.353	.574	.553	.015	.977	.392	.406
Tip-Adapter-F, biomedclip	.180	.670	.544	.149	.870	.326	.030	.912	.443	.805
WiSE-FT, siglip2_l16	.323	.675	.881	.537	.852	.197	.228	.617	.834	.928
Zero-shot, medsiglip	.679	.800	.689	.254	.556	.727	.727	.517	.586	.871

Table S12: ECE after TS per source, and the equal-source mean.

Method	Paraguay	HYGD	FIVES	PAPILA	PALM	HRF	AFIO	Chákṣu	Eq-src
LOTUS	.046	.084	.049	.058	.094	.301	.203	.091	.119
16-shot probe, medsiglip	.075	.103	.075	.321	.124	.175	.301	.060	.168
CoOp, siglip_b16	.106	.607	.119	.446	.137	.216	.388	.199	.288
CoOp ctx-init, biomedclip	.090	.640	.167	.242	.153	.295	.356	.170	.278
CoOp + LP, siglip_b16	.103	.611	.116	.443	.139	.234	.412	.203	.294
CORAL, convnext_t	.055	.444	.184	.385	.181	.291	.249	.206	.256
ERM, dinov2_lora	.044	.215	.305	.288	.113	.278	.234	.113	.211
Fine-tune, swin_t	.043	.480	.250	.411	.170	.331	.333	.239	.288
Fine-tune, NLL rule, convnext_t	.051	.423	.171	.346	.175	.337	.280	.211	.255
Group DRO, convnext_t	.051	.405	.156	.100	.149	.263	.299	.081	.203
Linear probe, medsiglip	.032	.184	.249	.063	.107	.315	.285	.140	.176
MLP probe, dinov2_b14	.046	.617	.179	.328	.130	.303	.313	.215	.274
SWAD, convnext_t	.051	.466	.166	.390	.199	.288	.242	.219	.258
Tip-Adapter, biomedclip	.070	.315	.065	.362	.184	.137	.085	.170	.174
Tip-Adapter-F, biomedclip	.053	.336	.109	.568	.178	.317	.315	.165	.268
WiSE-FT, siglip2_l16	.063	.442	.250	.484	.083	.338	.294	.084	.279
Zero-shot, medsiglip	.064	.133	.052	.132	.270	.192	.209	.214	.150

Table S13. LOTUS ECE after TS per source on the published split, three seeds pooled, with 95% bootstrap intervals over leakage groups (1,000 draws).

Source	Images	Groups	ECE (TS)	95% CI
Paraguay	1,431	1,271	.046	[.037, .062]
HYGD	548	183	.084	[.061, .123]
FIVES	586	574	.049	[.042, .084]
PAPILA	420	210	.058	[.050, .092]
PALM	161	161	.094	[.079, .152]
HRF	45	45	.301	[.196, .438]
AFIO	129	65	.203	[.157, .297]
Chákṣu	1,343	1,306	.091	[.078, .112]

Table S14: Per-source accuracy, bAcc, macro-F1 (over classes present) and ECE after TS, three seeds. *Single-class source: bAcc equals that class's recall and macro-F1 is degenerate.

Method	Source	n	Classes	Acc.	bAcc	Macro-F1	ECE (TS)
LOTUS	Paraguay	1,431	N/D	.920	.919	.936	.046
	HYGD*	548	G	.822	.822	.902	.084
	FIVES	586	N/D/G	.721	.721	.722	.049
	PAPILA	420	N/G	.802	.664	.712	.058
	PALM*	161	N	.836	.836	.911	.094
	HRF	45	N/D/G	.622	.622	.522	.301
	AFIO	129	N/D/G	.579	.652	.585	.203
	Chákṣu	1,343	N/G	.791	.695	.672	.091
16-shot probe, MedSigLIP	Paraguay	1,431	N/D	.875	.874	.902	.075
	HYGD*	548	G	.756	.756	.858	.103
	FIVES	586	N/D/G	.670	.668	.656	.075
	PAPILA	420	N/G	.288	.499	.298	.321
	PALM*	161	N	.723	.723	.836	.124
	HRF	45	N/D/G	.711	.711	.713	.175
	AFIO	129	N/D/G	.413	.548	.391	.301
	Chákṣu	1,343	N/G	.619	.675	.572	.060
Linear probe, MedSigLIP	Paraguay	1,431	N/D	.934	.934	.940	.032
	HYGD*	548	G	.477	.477	.646	.184
	FIVES	586	N/D/G	.561	.559	.532	.249
	PAPILA	420	N/G	.726	.493	.512	.063
	PALM*	161	N	.907	.907	.951	.107
	HRF	45	N/D/G	.637	.637	.531	.315
	AFIO	129	N/D/G	.509	.571	.447	.285
	Chákṣu	1,343	N/G	.590	.347	.392	.140
Zero-shot, MedSigLIP	Paraguay	1,431	N/D	.730	.729	.809	.064
	HYGD*	548	G	.719	.719	.837	.133
	FIVES	586	N/D/G	.724	.723	.723	.052
	PAPILA	420	N/G	.683	.622	.638	.132
	PALM*	161	N	.379	.379	.550	.270
	HRF	45	N/D/G	.556	.556	.511	.192
	AFIO	129	N/D/G	.457	.512	.469	.209
	Chákṣu	1,343	N/G	.406	.615	.396	.214
Group DRO, ConvNeXt-T	Paraguay	1,431	N/D	.920	.920	.924	.051
	HYGD*	548	G	.316	.316	.454	.405
	FIVES	586	N/D/G	.704	.704	.703	.156
	PAPILA	420	N/G	.683	.592	.614	.100
	PALM*	161	N	.870	.870	.928	.149
	HRF	45	N/D/G	.600	.600	.538	.263
	AFIO	129	N/D/G	.465	.539	.457	.299
	Chákṣu	1,343	N/G	.686	.417	.451	.081
ERM, DINOv2 LoRA	Paraguay	1,431	N/D	.885	.884	.897	.044
	HYGD*	548	G	.505	.505	.669	.215
	FIVES	586	N/D/G	.597	.596	.558	.305
	PAPILA	420	N/G	.448	.498	.483	.288
	PALM*	161	N	.816	.816	.898	.113
	HRF	45	N/D/G	.607	.607	.557	.278
	AFIO	129	N/D/G	.543	.614	.542	.234
	Chákṣu	1,343	N/G	.615	.419	.481	.113
Fine-tune, Swin-T	Paraguay	1,431	N/D	.913	.913	.916	.043
	HYGD*	548	G	.237	.237	.380	.480
	FIVES	586	N/D/G	.647	.647	.630	.250
	PAPILA	420	N/G	.380	.468	.426	.411
	PALM*	161	N	.836	.836	.910	.170
	HRF	45	N/D/G	.600	.600	.524	.331
	AFIO	129	N/D/G	.416	.501	.369	.333
	Chákṣu	1,343	N/G	.487	.298	.365	.239

Reading. DR is detected well by every method (AUROC 0.93–0.96). Glaucoma separates the methods: LOTUS reaches AUROC 0.926 on test against 0.79–0.87 for the baselines. On Chákṣu, fine-tuned and DG models detect almost no glaucoma (sensitivity 0.04–

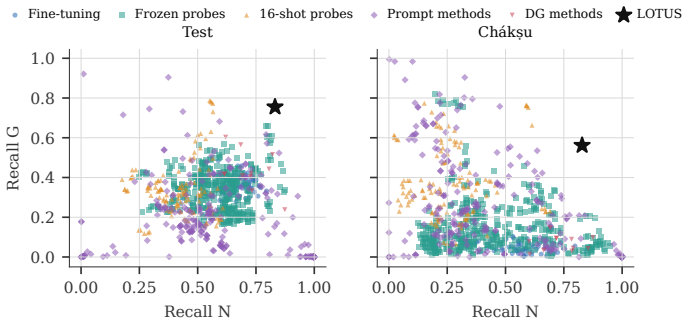


Fig. S1. Recall of normal against glaucoma images for every configuration. Few reach the upper right on either view: models either miss glaucoma or over-call it. LOTUS is among the most balanced on test; on Chákṣu it keeps normal recall high at the cost of glaucoma recall.

Table S15. Disease-level results, one disease against the rest, after TS, three seeds: sensitivity, specificity, positive predictive value and AUROC.

Method	View	Disease	Sens.	Spec.	PPV	AUROC
LOTUS	Test	D	.905	.931	.842	.961
LOTUS	Test	G	.755	.918	.770	.926
LOTUS	Chákṣu	G	.561	.866	.404	.841
16-shot probe, MedSigLIP	Test	D	.906	.873	.746	.957
16-shot probe, MedSigLIP	Test	G	.783	.767	.550	.870
16-shot probe, MedSigLIP	Chákṣu	G	.752	.703	.292	.803
Linear probe, MedSigLIP	Test	D	.896	.887	.762	.940
Linear probe, MedSigLIP	Test	G	.488	.908	.658	.854
Linear probe, MedSigLIP	Chákṣu	G	.011	.999	.600	.768
Zero-shot, MedSigLIP	Test	D	.850	.913	.798	.944
Zero-shot, MedSigLIP	Test	G	.682	.771	.521	.811
Zero-shot, MedSigLIP	Chákṣu	G	.904	.351	.184	.757
Group DRO, ConvNeXt-T	Test	D	.871	.848	.717	.929
Group DRO, ConvNeXt-T	Test	G	.443	.918	.658	.846
Group DRO, ConvNeXt-T	Chákṣu	G	.045	.992	.502	.676
ERM, DINOv2 LoRA	Test	D	.899	.840	.694	.929
ERM, DINOv2 LoRA	Test	G	.603	.835	.577	.831
ERM, DINOv2 LoRA	Chákṣu	G	.148	.947	.321	.650
Fine-tune, Swin-T	Test	D	.898	.819	.669	.927
Fine-tune, Swin-T	Test	G	.411	.869	.533	.789
Fine-tune, Swin-T	Chákṣu	G	.037	.973	.211	.622

0.15); LOTUS has the highest specificity among methods that detect a meaningful share of glaucoma (0.866, sensitivity 0.561, PPV 0.404), whereas the initial design reached sensitivity 0.822 at specificity 0.688. Precision for glaucoma on Chákṣu stays low for every method, because only 14% of its images are glaucoma; the higher values of the linear probe and Group DRO (0.50–0.60) come with sensitivities below 0.05.

Table S16. Sensitivity to label scope and single-class sources: test bAcc on all sources, without the single-class HYGD and PALM, and on the four sources that assessed normal images for glaucoma (AFIO, FIVES, HRF, PAPILA; 1,180 images).

Method	All test	Without single-class	Full-scope, multi-class
LOTUS	.830	.794	.728
16-shot probe, MedSigLIP	.748	.756	.608
Linear probe, MedSigLIP	.728	.730	.623
Zero-shot, MedSigLIP	.709	.698	.689
Group DRO, ConvNeXt-T	.707	.772	.673
ERM, DINOv2 LoRA	.707	.752	.605
Fine-tune, Swin-T	.655	.741	.591

Reading. On the four sources that graded every class, LOTUS remains first (0.728), ahead of zero-shot (0.689) and the probes (0.608–0.623); every method is lower there than on the pooled test, which Paraguay inflates.

Reading. Paraguay is easy for every method, whereas AFIO and HRF are hard for all. D transfers well everywhere; the differences between methods come mostly from G and from N on G-heavy sources. Per-source ECE after a single global temperature remains high for every method on HRF and AFIO, small sets where a pooled ECE hides the error. Table S13 shows that their intervals are wide (HRF 0.196–0.438 with 45 images), so calibration on sets this small cannot be certified either way; a deployment site needs a local calibration set of a few hundred images (Table S21).

S5 Reliability

Table S17: Selective prediction on test with thresholds fitted on Cal-B, raw/TS/FTS. Cov 80, Cov 90: thresholds for that coverage on Cal-B. Risk 10: SGR bound of 10%.

		Cov 80				Cov 90		Risk 10		Chákṣu Cov 80	
Method		Cov.	Risk	G→N	D→N	Cov.	Risk	Cov.	Risk	Cov.	Risk
LOTUS	R	.691	.083	.054	.035	.827	.114	.716	.087	.442	.064
	T	.691	.078	.054	.035	.830	.112	.732	.086	.437	.055
	F	.690	.078	.054	.035	.833	.111	.737	.087	.439	.051
16-shot probe, medsiglip	R	.757	.196	.012	.017	.883	.245	.354	.060	.528	.307
	T	.753	.195	.012	.016	.876	.241	.353	.057	.509	.307
	F	.752	.195	.012	.016	.876	.241	.357	.058	.504	.307
CoOp, siglip_b16	R	.801	.477	.039	.086	.898	.489	.000	–	.598	.648
	T	.797	.476	.038	.089	.898	.488	.000	–	.572	.649
	F	.798	.476	.038	.086	.899	.489	.000	–	.586	.647
CoOp ctx-init, biomedclip	R	.761	.454	.200	.068	.869	.469	.000	–	.808	.395
	T	.745	.453	.187	.065	.854	.468	.000	–	.811	.398
	F	.749	.452	.189	.067	.857	.468	.000	–	.812	.396
CoOp + LP, siglip_b16	R	.793	.470	.040	.085	.895	.484	.000	–	.572	.673
	T	.796	.470	.040	.090	.898	.484	.000	–	.551	.675
	F	.797	.469	.039	.088	.898	.484	.000	–	.559	.674

Table S17: (continued)

Method		Cov 80 Cov.	Cov 80 Risk	Cov 80 G→N	Cov 80 D→N	Cov 90 Cov.	Cov 90 Risk	Risk 10 Cov.	Risk 10 Risk	Chákṣu Cov 80 Cov.	Chákṣu Cov 80 Risk
CORAL, convnext_t	R	.668	.218	.157	.029	.807	.253	.727	.233	.530	.354
	T	.655	.212	.144	.028	.789	.245	.706	.224	.525	.353
	F	.668	.218	.158	.029	.807	.254	.726	.233	.529	.354
ERM, dinov2_lora	R	.726	.228	.075	.013	.848	.262	.697	.224	.566	.318
	T	.715	.223	.067	.012	.829	.255	.689	.221	.552	.314
	F	.723	.227	.073	.013	.839	.259	.695	.222	.561	.318
Fine-tune, swin_t	R	.708	.257	.156	.025	.819	.283	.766	.269	.457	.489
	T	.694	.250	.147	.024	.803	.276	.752	.263	.444	.484
	F	.709	.257	.156	.025	.819	.283	.766	.269	.458	.489
Fine-tune, NLL rule, convnext_t	R	.706	.219	.141	.033	.824	.253	.765	.237	.497	.423
	T	.690	.213	.126	.031	.809	.246	.755	.230	.489	.420
	F	.707	.220	.143	.033	.826	.254	.766	.238	.497	.423
Group DRO, convnext_t	R	.734	.180	.086	.042	.834	.208	.661	.162	.569	.249
	T	.738	.182	.089	.042	.838	.210	.664	.165	.570	.249
	F	.751	.187	.097	.044	.850	.215	.668	.167	.572	.249
Linear probe, medsiglip	R	.602	.127	.057	.022	.750	.161	.794	.176	.323	.271
	T	.610	.129	.061	.022	.765	.167	.800	.178	.308	.275
	F	.620	.132	.067	.023	.771	.169	.811	.183	.299	.276
MLP probe, dinov2_b14	R	.692	.290	.304	.048	.834	.335	.679	.288	.535	.461
	T	.672	.281	.281	.045	.812	.326	.666	.278	.524	.459
	F	.685	.287	.298	.047	.824	.330	.673	.285	.532	.462
SWAD, convnext_t	R	.708	.225	.172	.032	.832	.257	.820	.253	.478	.401
	T	.690	.216	.156	.030	.813	.246	.804	.245	.471	.399
	F	.704	.223	.167	.032	.825	.252	.816	.250	.475	.399
Tip-Adapter, biomedclip	R	.798	.549	.095	.153	.897	.554	.000	–	.794	.495
	T	.757	.547	.089	.113	.874	.549	.000	–	.930	.490
	F	.762	.546	.091	.116	.872	.548	.000	–	.923	.490
Tip-Adapter-F, biomedclip	R	.769	.526	.079	.045	.882	.526	.000	–	.810	.409
	T	.771	.524	.077	.046	.887	.525	.028	.446	.828	.416
	F	.770	.524	.077	.046	.887	.526	.030	.458	.828	.416
WiSE-FT, siglip2_l16	R	.659	.270	.100	.028	.804	.306	.679	.276	.430	.286
	T	.641	.260	.088	.025	.780	.295	.661	.268	.417	.280
	F	.654	.267	.097	.027	.798	.303	.673	.274	.426	.283
Zero-shot, medsiglip	R	.727	.230	.122	.052	.868	.270	.190	.040	.745	.581
	T	.711	.224	.107	.052	.861	.267	.182	.033	.689	.576
	F	.711	.224	.107	.052	.861	.267	.182	.033	.689	.576

Table S18: Conformal sets on test, quantiles from Cal-B. Coverage/mean size at nominal 0.90; coverage at nominal 0.95.

Method		LAC .90	APS .90	Class-LAC .90	LAC .95	APS .95	Class-LAC .95	Chákṣu LAC .90
LOTUS	R	.873/1.11	.939/1.72	.847/1.09	.954	.952	.942	.890/1.25
	T	.883/1.14	.955/1.88	.848/1.10	.956	.957	.941	.902/1.28
	F	.883/1.14	.928/1.70	.849/1.09	.955	.940	.941	.900/1.27
16-shot probe, medsiglip	R	.913/1.61	.923/1.99	.823/1.40	.966	.994	.919	.930/2.09
	T	.912/1.61	.922/1.98	.817/1.39	.967	.996	.916	.929/2.08
	F	.909/1.60	.925/2.01	.822/1.39	.966	.996	.922	.927/2.09
CoOp, siglip_b16	R	.761/1.80	.866/2.24	.761/1.83	.843	.947	.856	.817/2.26
	T	.760/1.80	.862/2.24	.760/1.83	.839	.944	.858	.811/2.23
	F	.760/1.79	.852/2.20	.773/1.85	.839	.986	.858	.802/2.18
CoOp ctx-init, biomedclip	R	.777/1.89	.762/1.91	.784/1.95	.846	.878	.845	.854/1.77
	T	.766/1.84	.760/1.90	.783/1.94	.835	.878	.846	.856/1.75
	F	.759/1.81	.750/1.87	.782/1.93	.825	.875	.850	.858/1.75
CoOp + LP, siglip_b16	R	.755/1.78	.809/2.01	.761/1.81	.839	.949	.847	.802/2.27
	T	.753/1.77	.808/2.00	.761/1.81	.837	.983	.849	.792/2.22
	F	.755/1.77	.795/1.96	.773/1.84	.832	.985	.853	.785/2.18
CORAL, convnext_t	R	.716/1.09	.815/1.68	.668/1.13	.780	.823	.754	.635/1.15
	T	.710/1.07	.807/1.60	.663/1.10	.773	.819	.749	.637/1.15
	F	.713/1.08	.796/1.56	.666/1.11	.769	.826	.741	.637/1.16
ERM, dinov2_lora	R	.728/1.10	.870/1.72	.697/1.13	.801	.885	.765	.684/1.17
	T	.722/1.08	.870/1.72	.689/1.11	.796	.884	.758	.679/1.15
	F	.723/1.08	.861/1.68	.692/1.11	.793	.882	.757	.679/1.15
Fine-tune, swin_t	R	.682/1.08	.794/1.65	.635/1.05	.737	.803	.720	.566/1.18
	T	.676/1.06	.786/1.59	.631/1.03	.732	.798	.715	.559/1.16
	F	.680/1.07	.784/1.67	.634/1.04	.730	.806	.714	.562/1.17
Fine-tune, NLL rule, convnext_t	R	.712/1.08	.829/1.73	.679/1.08	.771	.838	.769	.604/1.18
	T	.708/1.06	.828/1.71	.674/1.06	.764	.836	.762	.604/1.18
	F	.711/1.07	.806/1.58	.679/1.07	.763	.829	.757	.602/1.18
Group DRO, convnext_t	R	.767/1.13	.844/1.65	.733/1.09	.810	.854	.814	.787/1.29
	T	.769/1.13	.844/1.65	.733/1.09	.811	.854	.815	.788/1.29
	F	.771/1.14	.838/1.65	.735/1.10	.811	.855	.810	.790/1.30
Linear probe, medsiglip	R	.743/0.99	.883/1.68	.682/0.92	.857	.893	.765	.631/1.09
	T	.751/1.01	.888/1.72	.687/0.94	.863	.897	.765	.626/1.08
	F	.754/1.01	.855/1.55	.691/0.95	.857	.876	.760	.626/1.08
MLP probe, dinov2_b14	R	.658/1.11	.882/2.28	.637/1.13	.746	.931	.703	.576/1.17
	T	.651/1.09	.798/1.79	.629/1.11	.739	.807	.695	.573/1.16
	F	.652/1.09	.777/1.67	.633/1.12	.738	.798	.696	.573/1.17
SWAD, convnext_t	R	.698/1.04	.827/1.73	.668/1.05	.761	.834	.766	.562/1.09
	T	.690/1.01	.822/1.69	.662/1.02	.753	.831	.758	.558/1.08
	F	.693/1.02	.802/1.54	.666/1.03	.753	.825	.750	.560/1.09
Tip-Adapter, biomedclip	R	.947/2.50	.989/2.92	.785/2.00	.979	.989	.866	.994/2.27

Table S18: (continued)

Method		LAC .90	APS .90	Class-LAC .90	LAC .95	APS .95	Class-LAC .95	Chákṣu LAC .90
	T	.970/2.54	.993/2.95	.756/1.85	.988	.993	.845	1.00/2.24
	F	.961/2.52	.992/2.94	.764/1.90	.984	.992	.840	.997/2.25
Tip-Adapter-F, biomedclip	R	.819/2.09	.989/2.92	.764/1.88	.889	.989	.838	.897/2.09
	T	.825/2.09	.992/2.94	.766/1.86	.899	.992	.841	.901/2.10
	F	.817/2.08	.948/2.70	.763/1.87	.889	.990	.837	.898/2.10
WiSE-FT, siglip2_l16	R	.676/1.12	.805/1.78	.633/1.13	.738	.813	.707	.717/1.23
	T	.668/1.10	.804/1.77	.628/1.11	.732	.812	.700	.707/1.20
	F	.671/1.11	.790/1.70	.631/1.12	.729	.811	.701	.709/1.21
Zero-shot, medsiglip	R	.923/1.74	.896/1.80	.912/1.87	.981	.996	.967	.852/1.91
	T	.924/1.76	.920/1.89	.916/1.90	.982	.994	.968	.842/1.90
	F	.924/1.76	.920/1.89	.916/1.90	.982	.994	.968	.842/1.90

Reading. Methods whose sets stay near nominal coverage (16-shot, zero-shot) do so with mean set sizes of 1.6 to 1.8 out of three classes, which is of little use. Methods with compact sets lose coverage. TS keeps the ranking of confidence, so it barely changes coverage or risk at a fixed threshold.

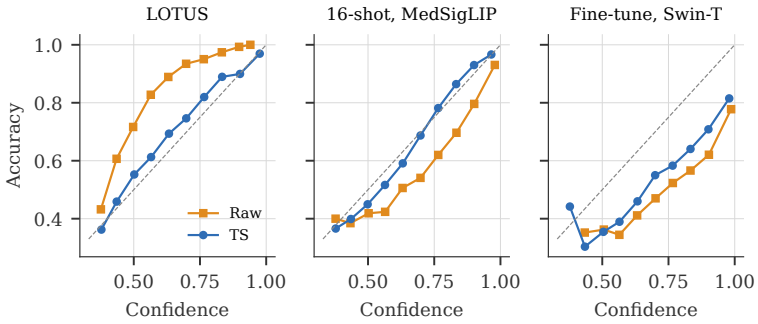


Fig. S2. Test reliability on the published split, seeds pooled, bins with at least 30 images. Raw LOTUS is underconfident and TS brings it near the diagonal; fine-tuned Swin-T stays overconfident after TS.

Reading. LOTUS has the lowest area under the risk–coverage curve of every method, on test (0.062; next: linear probe 0.110) and on Chákṣu (0.081; next: Group DRO 0.269). The Cal-B threshold lands at 69% coverage instead of 80%, and on Chákṣu at 44%: a threshold fitted on development data does not keep its coverage, and risk alone rewards models that call everything normal.

Reading. On the other half of Cal-B, the promises hold almost exactly; on test and Chákṣu they fail. The failure therefore comes from the new sources, not from a small calibration set.

Reading. The pooled risk target held (0.082), but per source it held on four of seven test sources and failed on AFIO, HRF, PALM and Chákṣu. A pooled guarantee is not a per-site guarantee, and neither is distribution-free under source shift; a deployment site must check its own promises (Table S37).

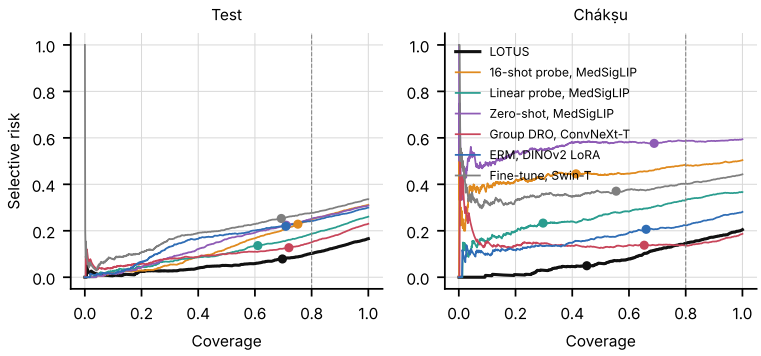


Fig. S3. Risk–coverage curves after TS on the published split (seed 0); dots mark the threshold fitted for 80% coverage on Cal-B. On Chákṣu, where 86% of images are normal, methods that rarely call glaucoma reach low risk while missing most disease.

Table S19. Is the failure caused by too few calibration images? Cal-B is split by leakage group 100 times; thresholds fit on one half are read on the other half (same sources), on test and on Chákṣu (TS).

	Coverage at Cov 80			Risk at Risk 10			LAC .90 coverage		
Method	Other half	Test	Chákṣu	Other half	Test	Chákṣu	Other half	Test	Chákṣu
16-shot probe, medsiglip	.799	.754	.507	.054	.041	.132	.900	.911	.926
CoOp, siglip_b16	.799	.797	.573	–	–	–	.901	.760	.811
CoOp ctx-init, biomedclip	.800	.744	.810	.134	.257	.294	.900	.767	.856
CoOp + LP, siglip_b16	.799	.795	.548	–	–	–	.901	.756	.796
CORAL, convnext_t	.797	.650	.522	.068	.209	.357	.901	.710	.632
ERM, dinov2_lora	.799	.714	.550	.068	.210	.289	.900	.722	.682
Fine-tune, swin_t	.800	.695	.446	.069	.251	.487	.901	.675	.554
Fine-tune, NLL rule, convnext_t	.798	.688	.487	.068	.213	.424	.902	.709	.605
Group DRO, convnext_t	.797	.737	.569	.064	.150	.248	.900	.771	.789
Linear probe, medsiglip	.800	.610	.308	.070	.161	.318	.901	.752	.632
MLP probe, dinov2_b14	.801	.673	.526	.069	.258	.452	.901	.651	.573
SWAD, convnext_t	.798	.689	.470	.070	.230	.423	.901	.691	.560
Tip-Adapter, biomedclip	.797	.757	.928	–	–	–	.900	.970	1.00
Tip-Adapter-F, biomedclip	.801	.772	.829	.084	.436	.642	.900	.827	.904
WiSE-FT, siglip2_l16	.799	.641	.420	.068	.258	.281	.900	.668	.708
Zero-shot, medsiglip	.799	.717	.696	.045	.027	.432	.902	.925	.844

Table S20. LOTUS's promises per unseen source on the published split (TS; thresholds fitted on Cal-B): coverage and risk at the thresholds for 80% and 90% coverage and at the SGR threshold for risk ≤ 10%, and whether that risk target held.

	Cov 80		Cov 90		Risk 10		
Source	Cov.	Risk	Cov.	Risk	Cov.	Risk	Held
AFIO	.450	.385	.646	.432	.509	.386	no
FIVES	.573	.128	.749	.192	.620	.135	no
HRF	.859	.345	.948	.367	.889	.358	no
HYGD	.581	.096	.793	.122	.642	.105	no
PALM	.584	.074	.758	.096	.650	.076	yes
PAPILA	.623	.071	.796	.131	.671	.088	yes
Paraguay	.829	.038	.909	.050	.854	.040	yes
Chákṣu	.437	.055	.696	.123	.511	.067	yes

Table S21: Calibration budget: fitted T on all of Cal-A, the oracle T fitted on test, and test NLL after TS with natural random subsets of Cal-A (median of 100 draws). $T = 1000$ marks the upper bound of the temperature search (zero-shot BiomedCLIP, whose calibrated output is then uniform, NLL = $\ln 3 \approx 1.10$).

Setting	Model	T	T oracle	NLL, 50	NLL, 200	NLL, 1000	NLL, all
16-shot probe	biomedclip	1.59	2.88	1.05	1.05	1.05	1.05
16-shot probe	clip_b16	1.80	3.28	1.08	1.08	1.07	1.07
16-shot probe	dinov2_b14	2.80	4.47	.997	.994	.993	.993
16-shot probe	resnet50	3.75	2.61	.979	.976	.977	.977
16-shot probe	siglip2_l16	1.57	1.75	.929	.926	.926	.926
CoOp	biomedclip	2.07	6.48	1.23	1.22	1.23	1.23
CoOp	clip_b16	1.43	4.17	1.21	1.20	1.19	1.20
CoOp	siglip2_l16	.780	.893	1.07	1.07	1.07	1.07
CoOp ctx-init	biomedclip	2.22	6.34	1.19	1.17	1.18	1.18
CoOp ctx-init	clip_b16	1.25	3.88	1.20	1.19	1.19	1.19
CoOp ctx-init	siglip2_l16	1.32	3.97	1.20	1.20	1.19	1.19
CoOp + LP	biomedclip	2.17	6.89	1.24	1.23	1.24	1.24
CoOp + LP	clip_b16	1.51	4.39	1.21	1.20	1.19	1.19
CoOp + LP	siglip2_l16	.358	.385	.898	.896	.896	.895
Fine-tune	resnet50	1.52	3.65	1.02	1.01	1.01	1.01
Fine-tune, NLL rule	resnet50	1.69	4.68	1.13	1.11	1.12	1.12
Linear probe	biomedclip	.866	1.79	1.06	1.04	1.04	1.04
Linear probe	clip_b16	.889	1.83	.971	.944	.951	.950
Linear probe	dinov2_b14	.999	2.04	.926	.891	.904	.901
Linear probe	resnet50	1.16	1.87	.924	.911	.911	.910
Linear probe	siglip2_l16	.999	2.22	1.01	.989	.993	.993
Multiple choice	Fundus-R1-7B	1.87	2.24	.821	.816	.817	.817
Multiple choice	Qwen2.5-VL-7B-Instruct	1.78	3.39	1.07	1.08	1.08	1.08
Multiple choice	llava-med-v1.5-mistral-7b-hf	.725	.483	1.07	1.07	1.07	1.07
Multiple choice	llava-onevision-qwen2-7b-ov-hf	1.96	11.32	1.30	1.28	1.28	1.28
MLP probe	biomedclip	1.74	13.47	1.90	1.85	1.85	1.85
MLP probe	clip_b16	2.36	16.03	1.99	1.97	1.95	1.94
MLP probe	dinov2_b14	2.09	7.61	1.32	1.25	1.26	1.26
MLP probe	siglip2_l16	1.91	11.20	1.78	1.75	1.74	1.73
Zero-shot	biomedclip	1000.00	137.44	1.10	1.10	1.10	1.10
Zero-shot	clip_b16	.785	2.12	1.15	1.15	1.15	1.15
Zero-shot	siglip2_l16	.604	1.69	1.17	1.18	1.17	1.17

Reading. About 200 calibration images give nearly the NLL of the full set, yet the fitted T stays far from the test oracle for several models: more calibration data from the same sources does not close the shift.

Table S22. Trivial predictors, the floor for every metric.

View	Predictor	Acc.	bAcc	Eq-src	ECE	NLL
Test	uniform	.445	.333	.429	.112	1.10
Test	train_prior	.445	.333	.429	.063	1.30
Test	always_N	.445	.333	.429	–	–
Test	always_D	.288	.333	.214	–	–
Test	always_G	.267	.333	.357	–	–
Chákṣu	uniform	.861	.500	.500	.527	1.10
Chákṣu	train_prior	.861	.500	.500	.353	.978
Chákṣu	always_N	.861	.500	.500	–	–
Chákṣu	always_D	.000	.000	.000	–	–
Chákṣu	always_G	.139	.500	.500	–	–

Table S23. Spearman correlation between validation and test bAcc across configurations within each setting. Settings with at most two configurations (the four DG methods and the NLL rule, 9 configurations) are not listed separately but are counted in the last row.

Setting	Configurations	Spearman
16-shot probe	130	.374
CoOp	70	.800
CoOp ctx-init	70	.927
CoOp + LP	50	.393
Fine-tune	65	.053
Linear probe	390	.344
MLP probe	11	.527
Tip-Adapter	3	.500
Tip-Adapter (acc. rule)	3	.500
Tip-Adapter-F	3	−.500
Tip-Adapter-F (acc. rule)	3	−.500
WiSE-FT	4	-1.00
Zero-shot	36	.831
All configurations	847	.781

Reading. A class-prior predictor reaches ECE 0.063 with chance bAcc, so low ECE alone says nothing about usefulness. The high overall rank correlation (0.781 here, over 847 rows that also count the accuracy-rule Tip-Adapter variants; 0.786 over the 816 configurations of the main paper) mostly reflects differences between families; within fine-tuning it is 0.053.

S6 Robustness Checks

Table S24. Fifty random 7/7 source rotations (published split excluded). Probes use class-weighted cross-entropy; arrows go from internal estimates on held-back Cal-B to unseen test sources, after TS.

Method	Val	Test	5–95%	Eq-src	Rec. G	ECE raw	ECE TS	Risk at Cov 80	LAC .90	Coverage .80
LOTUS	.876	.788	.721–.842	.766	.708	.161	.034	.065→.101	.900→.848	.795→.722
medsiglip, full	.922	.745	.621–.827	.725	.637	.095	.084	.042→.134	.901→.769	.795→.668
medsiglip, 16-shot	.814	.716	.626–.817	.713	.723	.123	.056	.164→.217	.899→.868	.803→.778
dinov2_b14, full	.892	.655	.529–.765	.650	.531	.123	.118	.077→.221	.899→.735	.794→.659
dinov2_b14, 16-shot	.713	.552	.432–.659	.571	.505	.192	.073	.296→.383	.900→.860	.801→.779
siglip2_l16, full	.876	.631	.549–.731	.617	.549	.155	.141	.102→.280	.900→.728	.804→.680
siglip2_l16, 16-shot	.722	.578	.476–.683	.562	.586	.160	.080	.292→.372	.901→.871	.796→.751
convnext_t, full	.860	.593	.483–.700	.609	.426	.149	.130	.112→.287	.901→.730	.800→.684
convnext_t, 16-shot	.679	.494	.379–.604	.504	.456	.223	.091	.352→.463	.900→.840	.795→.771

Reading. Validation exceeded test in 98% of the 400 probe rows, risk exceeded its internal estimate in 89%, and LAC fell below its internal coverage in 85% (338 of 400); for LOTUS the shares were 98%, 86% and 92%. Ranking is not stable either: the 16-shot probe, best on the published split among representatives, is below full training on every encoder over the rotations.

Table S25. Full factorial ablation over the 50 rotations: every combination of bag, between-site projection and text, and text alone; TS, means over three seeds. Last two columns: LOTUS minus the row in bAcc, and rotations where LOTUS is higher.

Configuration	bAcc	Macro-F1	AUROC	Rec. G	False G	ECE	NLL	AURC	Risk	Chákṣu	ΔbAcc	Higher
LOTUS (bag + text)	.789	.739	.931	.709	.070	.034	.507	.071	.101	.694	–	–
single head + text	.786	.737	.930	.695	.069	.035	.509	.071	.099	.688	+0.003	35/50
bag only	.779	.720	.928	.725	.088	.050	.566	.090	.106	.598	+0.010	27/50
+ projection (initial design)	.771	.718	.921	.666	.081	.033	.526	.075	.117	.668	+0.018	45/50
single head	.770	.713	.924	.699	.086	.053	.577	.093	.106	.590	+0.019	35/50
single head + projection + text	.768	.715	.920	.656	.080	.033	.530	.076	.116	.663	+0.021	47/50
bag + projection	.754	.694	.911	.657	.099	.055	.600	.094	.125	.624	+0.035	48/50
single head + projection	.745	.686	.904	.637	.101	.058	.618	.100	.130	.612	+0.044	49/50
text only	.713	.653	.886	.614	.131	.022	.611	.109	.190	.594	+0.076	50/50

Reading. Every arm with the projection is below the same arm without it (bag + text: 0.789 against 0.771; single head + text: 0.786 against 0.768; bag: 0.779 against 0.754; single head: 0.770 against 0.745), so the projection's cost is consistent rather than an interaction. Text lowers ECE and NLL in every combination; the bag adds little once text is present (0.003) but more without it (0.009 for the single head).

Table S26. Three training seeds per rotation (rotations where all three finished; the probe uses class-weighted cross-entropy as in Table S24). bAcc: mean ± SD of seed-averaged bAcc across rotations; seed SD: SD across seeds within a rotation, averaged. Last columns: LOTUS minus the row, mean ± SD over rotations, and rotations where LOTUS has higher bAcc and lower ECE.

Configuration	Rot.	bAcc	Seed SD	ECE	Rec. G	ΔbAcc	Higher	Lower ECE
LOTUS	50	.789 ± .039	.004	.034	.709	–	–	–
+ projection (initial design)	50	.771 ± .039	.003	.033	.666	+0.018 ± 0.015	45/50	23/50
single head + text	50	.786 ± .041	.005	.035	.695	+0.003 ± 0.007	35/50	33/50
bag only	50	.779 ± .050	.007	.050	.725	+0.010 ± 0.028	27/50	34/50
text only	50	.713 ± .036	.000	.022	.614	+0.076 ± 0.027	50/50	11/50
MedSigLIP linear probe	50	.746 ± .069	.010	.084	.642	+0.043 ± 0.049	43/50	47/50

Reading. Seed variation is an order of magnitude smaller than variation between source partitions, so single-seed rotations measure the partition effect. Against the probe, LOTUS has higher bAcc on 43 and lower ECE on 47 of 50 rotations; text only is better calibrated (lower ECE than LOTUS on 39 of 50) but far less accurate.

Reading. Fitted on all development data ($k = 14$), the between-site directions hold 79–87% of every class-mean difference; for the individual heads ($k = 11$–13) they hold 55–87%, against 15–23% for a random subspace of equal size. Sites differ in severity and case mix within each label, so their within-class differences are themselves disease directions; projecting them out removes signal, which explains the accuracy loss over the rotations.

Reading. The source is almost perfectly decodable from every representation, frozen or fine-tuned (0.93–1.00), even within one class. Fine-tuning does not add site information,

Table S27. Why projection hurts. Share of the norm of each class-mean difference that lies in the removed between-site subspace (k directions), on the development data used by each head, against a random subspace of the same size; and alignment between the two glaucoma sites' difference and the N–G direction.

Source left out	k	N–D	N–G	D–G	N–G, random	| cos | G-site, N–G
none (all)	14	.868	.788	.811	.187	.098
Bangladesh Eye	11	.839	.548	.602	.225	–
OIA-DDR	12	.830	.839	.817	.208	.073
DeepDRiD	12	.850	.776	.762	.208	.083
GRAPE	13	.866	.819	.836	.147	–
IDRiD	12	.818	.784	.793	.209	.100
Jichi	12	.832	.726	.747	.218	.048
SUSTech-SYSU	12	.791	.785	.782	.207	.115

Table S28. How much of the source can be read from each representation. Five-fold cross-validated balanced accuracy of a logistic probe (C = 0.05) predicting the disease, the source, or the source within normal or DR images only, on development (validation and calibration) rows and on test rows. Chance for the source is 1/6 or 1/7.

Features	Rows	n	Disease	Source	Source within N	Source within D
Fine-tuned Swin-T	development	5,951	.891	.951	.963	.931
Fine-tuned Swin-T	test	3,320	.908	.977	.996	.996
Group DRO ConvNeXt-T	development	5,951	.867	.930	.933	.909
Group DRO ConvNeXt-T	test	3,320	.906	.980	.994	.944
ERM DINOv2 LoRA	development	5,951	.873	.946	.967	.914
ERM DINOv2 LoRA	test	3,320	.911	.986	.998	.995
Frozen MedSigLIP (pooled)	development	5,951	.895	.961	.949	.949
Frozen MedSigLIP (pooled)	test	3,320	.931	.998	.999	.986
Frozen MedSigLIP tokens (LOTUS input)	development	5,951	.890	.962	.958	.959
Frozen MedSigLIP tokens (LOTUS input)	test	3,320	.924	.998	.999	.986
Tokens after projection (initial design)	development	5,951	.855	.354	.318	.322
Tokens after projection (initial design)	test	3,320	.924	.998	.999	.984

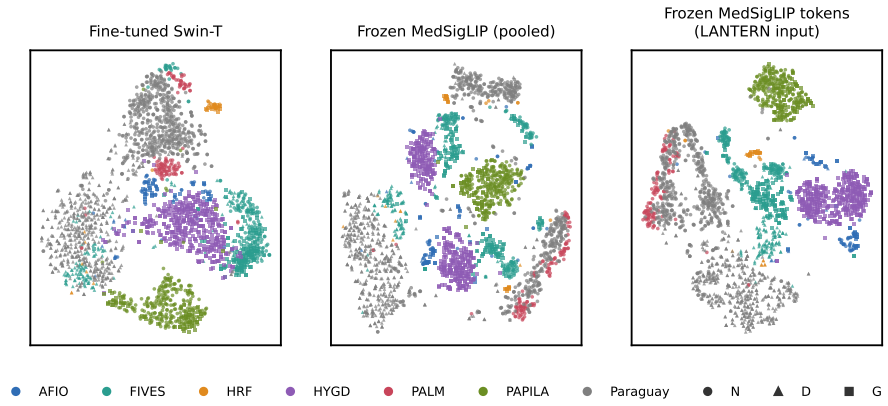


Fig. S4. t-SNE of test images (2,500 sampled), coloured by source, marker by class, for a fine-tuned encoder and the two frozen MedSigLIP representations. Every representation clusters by source.

so its failure lies in what its classifier relies on, not in what its features contain; frozen features carry the same site signal, and a low-capacity head simply exploits it less. The between-site projection removes site information for the development sources (0.962 to 0.354) but not for unseen ones (0.998), while it costs disease signal (0.890 to 0.855): it removes the known sites' directions, and new sites differ along new ones.

Table S29. Glaucoma recall over the 50 rotations by the number of development sources that contain glaucoma.

Method	G sources in development	Rotations	Recall G	Test bAcc
LOTUS	2	8	.614	.777
LOTUS	3	18	.718	.793
LOTUS	4	17	.723	.789
LOTUS	5	6	.756	.788
LOTUS	6	1	.777	.821
Initial design	2	8	.555	.761
Initial design	3	18	.671	.774
Initial design	4	17	.702	.774
Initial design	5	6	.684	.762
Initial design	6	1	.748	.801
MedSigLIP LP	2	8	.468	.676
MedSigLIP LP	3	18	.639	.742
MedSigLIP LP	4	17	.674	.766
MedSigLIP LP	5	6	.737	.772
MedSigLIP LP	6	1	.940	.878

Reading. Glaucoma recall grows with the number of glaucoma sources on the development side, so the weakness is one of site diversity, which more sources can address. LOTUS stays ahead of the linear probe at every diversity level observed in more than one rotation (two sources: 0.777 against 0.676; five: 0.788 against 0.772), whereas the initial design fell behind the probe with five sources (0.762 against 0.772).

Reading. Region hold-out is harsher than the rotations for some regions: holding out East Asia removes GRAPE, leaving one glaucoma development source. LOTUS has the highest mean bAcc (0.822), clearly ahead on South Asia (0.751 against 0.678) and the Middle East, within 0.013 of the probe elsewhere, and the best Chákṣu bAcc in every hold-out; its calibration is not better than the probe's here (mean ECE 0.061 against 0.058).

Reading. N–D and D–G stay near 0.90–0.95 while N–G is 0.64–0.75, so glaucoma is a ranking failure, not a threshold failure. Within-source learning recovers much of it (Chákṣu gap +0.372), so the labels are learnable and the loss is tied to the shift. AFIO stays low even within source.

Reading. More sources usually help on test (MedSigLIP 0.594 to 0.817 from one to seven sources) but not on Chákṣu (0.534 to 0.350), and more labels from the same sources give little after about 64 per class. Shift is not solved by more of the same data.

Table S30. Leave-one-region-out: all non-OOD sources of one region form the test side, the rest the development side (80/10/10 group split), three seeds. Eq-src: mean per-source bAcc; Chákṣu is read for every split.

Region held out	Test sources	Images	Method	bAcc	Eq-src	Rec. G	ECE	Chákṣu
East Asia	OIA-DDR, DeepDRiD, FIVES, GRAPE, Jichi, PALM, SUSTech-SYSU	26,487	**LOTUS**	.810	.820	.822	.058	.728
			Initial design	.773	.777	.743	.047	.650
			MedSigLIP LP	.810	.822	.815	.055	.559
Europe	HRF, PAPILA	465	**LOTUS**	.726	.646	.448	.051	.705
			Initial design	.740	.662	.657	.057	.680
			MedSigLIP LP	.739	.662	.497	.055	.562
Latin America	Paraguay	1,431	**LOTUS**	.913	.913	–	.044	.726
			Initial design	.862	.862	–	.021	.699
			MedSigLIP LP	.917	.917	–	.031	.527
Middle East	HYGD	548	**LOTUS**	.909	.909	.909	.104	.715
			Initial design	.850	.850	.850	.068	.689
			MedSigLIP LP	.898	.898	.898	.061	.548
South Asia	AFIO, Bangladesh Eye, IDRiD	4,145	**LOTUS**	.751	.714	.563	.046	.691
			Initial design	.729	.687	.438	.075	.638
			MedSigLIP LP	.678	.702	.443	.087	.601
Mean			**LOTUS**	.822	.800	–	.061	–
Mean			Initial design	.791	.768	–	.054	–
Mean			MedSigLIP LP	.808	.800	–	.058	–

Table S31. Pairwise AUROC, mean over published test sources holding both classes, with bootstrap intervals over sources.

Pair	A	B	Sources	AUROC A	AUROC B	A−B	95% CI
D–G	16-shot probe	Fine-tune	3	.939	.916	+0.023	[-0.029, +0.077]
D–G	16-shot probe	Linear probe	3	.939	.917	+0.022	[-0.012, +0.059]
D–G	16-shot probe	Group DRO	3	.939	.942	-0.003	[-0.035, +0.033]
D–G	Linear probe	Fine-tune	3	.917	.916	+0.001	[-0.062, +0.081]
D–G	Group DRO	ERM	3	.942	.916	+0.025	[+0.010, +0.041]
N–D	16-shot probe	Fine-tune	4	.946	.900	+0.046	[+0.016, +0.074]
N–D	16-shot probe	Linear probe	4	.946	.928	+0.019	[-0.005, +0.053]
N–D	16-shot probe	Group DRO	4	.946	.923	+0.024	[+0.007, +0.045]
N–D	Linear probe	Fine-tune	4	.928	.900	+0.028	[-0.012, +0.069]
N–D	Group DRO	ERM	4	.923	.909	+0.013	[-0.021, +0.054]
N–G	16-shot probe	Fine-tune	4	.749	.643	+0.106	[+0.038, +0.203]
N–G	16-shot probe	Linear probe	4	.749	.678	+0.071	[-0.011, +0.152]
N–G	16-shot probe	Group DRO	4	.749	.704	+0.045	[-0.006, +0.118]
N–G	Linear probe	Fine-tune	4	.678	.643	+0.035	[-0.059, +0.135]
N–G	Group DRO	ERM	4	.704	.644	+0.060	[-0.015, +0.123]

Table S32. Trained on development sources against trained within the target (logistic regression on frozen features, mean over nine encoders; within-source by cross-validation). Gap: within minus development bAcc, with its bootstrap interval.

	bAcc		ECE		AUROC N–G		AUROC N–D		
Source	Dev	Within	Dev (TS)	Within	Dev	Within	Dev	Within	Gap
AFIO	.516	.635	.344	.153	.603	.745	.856	.829	+0.126 [-0.001, +0.249]
Chákṣu	.352	.718	.261	.044	.638	.806	–	–	+0.372 [+0.328, +0.415]
FIVES	.581	.857	.227	.049	.780	.961	.883	.969	+0.283 [+0.245, +0.320]
HRF	.583	.779	.295	.182	.477	.854	.926	.927	+0.193 [+0.053, +0.328]
PAPILA	.394	.757	.212	.070	.635	.832	–	–	+0.371 [+0.296, +0.445]
Paraguay	.817	.928	.046	.018	–	–	.934	.975	+0.115 [+0.095, +0.137]

Table S33. More training sources at a fixed size (300 images per class): test and Chákṣu bAcc for $K = 1, 2, 4, 7$ development sources (mean over random pools).

	Test				Chákṣu			
Encoder	1	2	4	7	1	2	4	7
clip_b16	.509	.478	.508	.558	.527	.446	.431	.494
convnext_t	.481	.514	.518	.557	.514	.491	.408	.440
dinov2_b14	.544	.618	.593	.720	.535	.555	.485	.543
flair†	.555	.633	.678	.758	.511	.624	.643	.679
medsiglip	.594	.725	.743	.817	.534	.561	.504	.350
resnet50	.547	.544	.503	.628	.564	.552	.362	.455
siglip2_l16	.512	.518	.548	.602	.494	.456	.387	.102
vit_b16_in21k	.417	.478	.542	.576	.504	.456	.347	.430

Table S34. Learning curve: change in test bAcc at k labels per class against the next smaller budget, with bootstrap intervals.

Encoder	k	ΔbAcc	95% CI
medsiglip	16	-0.008	[-0.161, +0.130]
medsiglip	64	+0.021	[-0.111, +0.124]
medsiglip	128	+0.040	[-0.060, +0.120]
medsiglip	512	+0.009	[-0.054, +0.078]
dinov2_b14	16	-0.130	[-0.257, +0.032]
dinov2_b14	64	-0.044	[-0.151, +0.088]
dinov2_b14	128	-0.016	[-0.114, +0.089]
dinov2_b14	512	+0.003	[-0.076, +0.060]
siglip2_l16	16	-0.050	[-0.129, +0.024]
siglip2_l16	64	-0.015	[-0.096, +0.087]
siglip2_l16	128	+0.025	[-0.080, +0.125]
siglip2_l16	512	+0.007	[-0.103, +0.056]

S7 Attempted Fixes and Other Probes

Table S35. Test-time adaptation per held-out source: test bAcc for none, BatchNorm statistics, TENT and T3A (K = 20, 100); Chákṣu bAcc for none, TENT and T3A-100.

Setting	Model	None	BN	TENT	T3A-20	T3A-100	Ch. none	Ch. TENT	Ch. T3A
Fine-tune	swin_t	.655	–	.655	.666	.668	.298	.297	.309
Fine-tune	resnet50	.661	.598	.598	.632	.641	.417	.363	.306
Group DRO	convnext_t	.707	–	.709	.705	.703	.417	.430	.358
ERM	dinov2_lora	.708	–	.710	.644	.662	.420	.421	.298
16-shot probe	medsiglip	.748	–	–	.659	.683	.675	–	.496
Linear probe	medsiglip	.728	–	–	.679	.679	.347	–	.271

Table S36. Prior correction on top of TS: test bAcc with EM (Saerens et al.) pooled or per source, and with true priors pooled or per source (oracle, uses test labels); ECE for TS, EM per source and oracle per source.

Setting	Model	TS	EM pooled	EM per src.	True pooled	Oracle per src.	ECE TS	ECE EM	ECE oracle
Zero-shot	medsiglip	.709	.540	.635	.709	.863	.032	.287	.033
CoOp	siglip_b16	.491	.398	.457	.513	.737	.175	.433	.050
CoOp ctx-init	biomedclip	.497	.416	.438	.495	.754	.150	.389	.040
CoOp + LP	siglip_b16	.496	.394	.450	.520	.736	.176	.445	.053
16-shot probe	medsiglip	.748	.683	.787	.769	.853	.044	.185	.051
Linear probe	medsiglip	.728	.778	.835	.778	.853	.079	.120	.081
MLP probe	dinov2_b14	.598	.610	.618	.621	.821	.172	.226	.070
Tip-Adapter-F (acc. rule)	biomedclip	.495	.407	.434	.511	.739	.147	.389	.047
Tip-Adapter (acc. rule)	biomedclip	.478	.333	.495	.460	.779	.091	.292	.120
Tip-Adapter-F	biomedclip	.495	.407	.434	.511	.739	.147	.389	.047
Tip-Adapter	biomedclip	.478	.333	.495	.460	.779	.091	.292	.120
Fine-tune	swin_t	.655	.657	.661	.675	.827	.171	.226	.099
Fine-tune, NLL rule	convnext_t	.677	.687	.710	.707	.842	.133	.182	.073
CORAL	convnext_t	.676	.687	.704	.704	.851	.136	.172	.069
SWAD	convnext_t	.672	.691	.704	.706	.851	.139	.166	.060
Group DRO	convnext_t	.707	.720	.723	.733	.857	.091	.188	.051
ERM	dinov2_lora	.707	.655	.726	.727	.826	.115	.218	.104
WiSE-FT	siglip2_l16	.634	.598	.627	.669	.813	.168	.281	.103

Reading. No label-free fix helps reliably. TENT changes little, BatchNorm adaptation harms ResNet-50 (notably on single-class PALM), and T3A is mostly negative. EM prior estimation is unstable and usually worsens ECE. True per-source priors raise bAcc to 0.81–0.86 for most settings, so class balance is a large part of the shift, but this oracle uses test labels, and single-class sources make it easy.

Table S37: K labelled images from each target source, read on the rest of that source: ECE with the frozen T against T re-fitted on K; LAC .90 coverage with the frozen quantile against a quantile from K, with the 5th percentile over draws.

Setting	Model	K	ECE frozen	ECE T from K	LAC frozen	LAC from K	LAC 5%
16-shot probe	medsiglip	10	.156	.128	.879	.897	.714
		25	.152	.093	.869	.920	.801

Table S37: (continued)

Setting	Model	K	ECE frozen	ECE T from K	LAC frozen	LAC from K	LAC 5%
		50	.154	.087	.869	.899	.808
		100	.129	.067	.900	.901	.837
Linear probe	medsiglip	10	.172	.134	.677	.899	.704
		25	.153	.098	.682	.917	.785
		50	.154	.086	.682	.898	.797
		100	.134	.073	.672	.901	.841
Zero-shot	medsiglip	10	.163	.136	.884	.901	.722
		25	.155	.103	.884	.920	.809
		50	.156	.094	.884	.900	.813
		100	.121	.080	.919	.902	.843
ERM	dinov2_lora	10	.200	.140	.658	.904	.714
		25	.189	.110	.663	.920	.819
		50	.191	.104	.663	.899	.805
		100	.194	.089	.643	.901	.831
Group DRO	convnext_t	10	.189	.140	.712	.903	.715
		25	.178	.117	.721	.916	.803
		50	.180	.109	.721	.899	.800
		100	.160	.101	.718	.901	.837
Fine-tune	swin_t	10	.283	.125	.590	.904	.712
		25	.276	.095	.588	.921	.809
		50	.277	.088	.588	.901	.811
		100	.285	.066	.557	.902	.838

Reading. Ten labels from the new source already bring LAC coverage near 0.90 on average and re-fitted T lowers ECE, but the worst draws stay far below nominal at small K; this is a deployment recipe, not a guarantee.

Reading. CLAHE does not help fine-tuning or the 16-shot probe. The MedSigLIP linear probe rises on test, but its validation falls, so validation-based selection would not adopt it.

Reading. MSP accepts a third to two thirds of out-of-scope images. KNN rejects most of them and lowers risk, but largely by dropping whole hard sources; it is useful for flagging strange images rather than as a remedy for shift.

Reading. Generative multimodal models rarely call glaucoma. Mild DR (grade 1) is the hardest D grade for most models. Strong blur, and for most models JPEG compression, mainly remove glaucoma recall, consistent with the disc being a small, detail-dependent region; the 16-shot MedSigLIP probe is the exception under JPEG (0.784 to 0.870).

S8 LOTUS: Specification and Development

S8.1 Design search on development sources

The design search below produced the initial design; LOTUS removes its projection after the rotation ablation (Table S25). All results in this section are leave-one-development-source-out on development data, equal-site mean, three seeds unless noted, and use no test row. Table S45 compares losses (cross-entropy; a conditional-logit pair loss across sites, "cross"; the same within sites, "strata"), a single head against a leave-one-source-out bag, the between-site projection ("nap"), and text fusion with one bank or all three, on pooled MedSigLIP features.

Table S38. CLAHE against unprocessed images, same seeds and hardware.

Setting	Config	Preprocessing	Val	Test	Eq-src	Chákṣu	ECE TS	Rec. G
Fine-tune	convnext_t_ce	none	.886	.668	.612	.273	.133	.385
Fine-tune	convnext_t_ce	CLAHE	.876	.653	.604	.234	.132	.293
Fine-tune	swin_t_mbls	none	.893	.655	.600	.298	.171	.411
Fine-tune	swin_t_mbls	CLAHE	.882	.650	.608	.283	.152	.351
Linear probe	medsiglip_class_ls	none	.920	.728	.654	.347	.079	.488
Linear probe	medsiglip_class_ls	CLAHE	.911	.788	.702	.400	.041	.663
16-shot probe	medsiglip_none_ce	none	.802	.748	.683	.675	.044	.783
16-shot probe	medsiglip_none_ce	CLAHE	.790	.747	.693	.678	.033	.804

Table S39. Referral rules at a threshold for 80% coverage on Cal-B: maximum softmax (MSP), KNN distance in feature space, or both. Open: share of the 861 out-of-scope images accepted.

Setting	Model	Score	Test cov.	Test risk	Cal-B cov.	Open	G→N
Fine-tune	swin_t	MSP	.694	.250	.801	.666	.147
Fine-tune	swin_t	KNN	.415	.084	.801	.165	.007
Fine-tune	swin_t	MSP+KNN	.484	.113	.802	.252	.020
Group DRO	convnext_t	MSP	.738	.182	.801	.602	.089
Group DRO	convnext_t	KNN	.400	.076	.801	.101	.010
Group DRO	convnext_t	MSP+KNN	.487	.094	.802	.187	.026
ERM	dinov2_lora	MSP	.715	.223	.801	.610	.067
ERM	dinov2_lora	KNN	.549	.266	.801	.216	.113
ERM	dinov2_lora	MSP+KNN	.628	.221	.804	.290	.092
Linear probe	medsiglip	MSP	.610	.129	.801	.337	.061
Linear probe	medsiglip	KNN	.419	.227	.801	.059	.093
Linear probe	medsiglip	MSP+KNN	.421	.153	.802	.074	.057
16-shot probe	medsiglip	MSP	.753	.195	.801	.679	.012
16-shot probe	medsiglip	KNN	.419	.223	.801	.059	.027
16-shot probe	medsiglip	MSP+KNN	.483	.183	.802	.105	.020

Table S40. Out-of-scope detection AUROC from penultimate features of trained models (SWAD rows omitted: its saved checkpoint holds pre-averaging weights).

Setting	Model	MSP	MaxLogit	Energy	KNN	Mahalanobis
Fine-tune	swin_t	.549	.567	.572	.664	.754
Group DRO	convnext_t	.618	.610	.606	.745	.791
ERM	dinov2_lora	.613	.610	.610	.753	.754

Table S41. Multiple-choice answering by multimodal LLMs (mean over six option orders).

Model	Test bAcc	Rec. N	Rec. D	Rec. G	Chákṣu bAcc	Chákṣu Rec. G
Fundus-R1-7B	.576	.789	.795	.143	.463	.128
Qwen2.5-VL-7B-Instruct	.424	.926	.347	.000	.488	.005
llava-med-v1.5-mistral-7b-hf	.347	.905	.056	.080	.507	.037
llava-onevision-qwen2-7b-ov-hf	.407	.367	.855	.000	.096	.000

Table S42. Recall of D by DR grade on test (grade −1: Davis grading without a numeric ICDR grade).

Setting	Model	−1	1	2	3	4
Zero-shot	medsiglip	.785	.000	.734	.954	.838
CoOp	siglip_b16	.711	.444	.688	.823	.915
CoOp ctx-init	biomedclip	.808	.111	.618	.884	.925
CoOp + LP	siglip_b16	.718	.444	.700	.835	.929
16-shot probe	medsiglip	.767	.056	.841	.989	.967
Linear probe	medsiglip	.727	.000	.844	.981	.977
MLP probe	dinov2_b14	.665	.222	.737	.935	.932
Tip-Adapter-F (acc. rule)	biomedclip	.660	.167	.569	.831	.885
Tip-Adapter (acc. rule)	biomedclip	.368	.000	.122	.432	.499
Tip-Adapter-F	biomedclip	.660	.167	.569	.831	.885
Tip-Adapter	biomedclip	.368	.000	.122	.432	.499
Fine-tune	swin_t	.724	.389	.844	.974	.986
Fine-tune, NLL rule	convnext_t	.717	.389	.801	.965	.969
CORAL	convnext_t	.704	.278	.801	.966	.972
SWAD	convnext_t	.724	.278	.817	.966	.972
Group DRO	convnext_t	.712	.111	.743	.969	.952
ERM	dinov2_lora	.701	.333	.865	.980	.996
WiSE-FT	siglip2_l16	.695	.222	.755	.964	.968

Table S43. Glaucoma recall on clean test images and at the strongest blur and JPEG levels.

Setting	Model	Clean	Blur	JPEG
Fine-tune	swin_t	.419	.071	.100
Group DRO	convnext_t	.447	.026	.073
ERM	dinov2_lora	.598	.438	.358
16-shot probe	medsiglip	.784	.267	.870
Linear probe	medsiglip	.491	.144	.086

Table S44. Complete specification of LOTUS. The initial design adds the projection row before the heads.

Encoder	MedSigLIP (patch 14, 448 px, 27 layers, width 1152), frozen
Head input	Last hidden state, 32×32 tokens pooled to 14×14; fixed random orthonormal map to 384 dimensions (seed 0); grid mean
Standardization	Per dimension, training-split mean and standard deviation
Projection	None. Initial design only: within-class between-site means, at most 32 directions, fitted per head on its training rows
Heads	One linear head per development source, trained without it; 1,155 parameters each
Training	Label smoothing 0.1; class-balanced sampling; AdamW, lr 3×10^{-4}, wd 10^{-2}; batch 64; 12 epochs; best epoch by validation bAcc on the head's own sites
Rescaling	Centre per image; divide by the standard deviation over validation rows
Text	Mean of the three banks of Table S8, on the pooled image embedding
Fusion	0.5 heads + 0.5 text; TS on Cal-A
Seeds	0, 1, 2 on the published split and on every rotation; the main rotation table uses seed 0 for all methods

Table S45: Factorial comparison on pooled features (development, three seeds).

Arm	bAcc	ECE	NLL	Recall G
cross bag nap+allbanks	.794	.156	.574	.636
strata bag nap+allbanks	.792	.151	.573	.625
ce bag nap+allbanks	.791	.155	.581	.639
strata bag+allbanks	.784	.148	.584	.618
cross bag nap+text	.784	.137	.577	.633
cross bag+allbanks	.782	.149	.594	.557
strata single nap+allbanks	.782	.144	.578	.692
strata bag nap+text	.781	.131	.577	.630
ce bag nap+text	.780	.134	.584	.640
ce bag+allbanks	.777	.149	.604	.576
cross single nap+allbanks	.776	.141	.589	.574
strata bag+text	.774	.129	.589	.622
strata single nap+text	.773	.128	.583	.696
cross bag+text	.773	.134	.597	.570
cross single nap+text	.770	.125	.593	.589
strata single+allbanks	.770	.135	.595	.594
ce bag+text	.768	.132	.606	.589
ce single nap+allbanks	.765	.136	.601	.583
strata single+text	.762	.120	.599	.607
cross bag nap	.762	.133	.623	.625
cross single+allbanks	.760	.135	.616	.472
strata bag nap	.759	.127	.621	.603
ce single nap+text	.758	.117	.604	.585
cross single+text	.753	.123	.619	.485
ce single+allbanks	.752	.143	.622	.531
ce bag nap	.750	.129	.641	.629
strata bag	.747	.130	.640	.597
ce single+text	.746	.130	.624	.541
cross bag	.737	.146	.663	.481
strata single nap	.731	.107	.651	.712
cross single nap	.729	.126	.666	.526
strata single	.723	.129	.675	.565
ce bag	.721	.145	.687	.507
ce single nap	.704	.125	.694	.517
cross single	.691	.158	.715	.319
ce single	.675	.169	.741	.417

Reading. Every component adds a little alone; with all three in place the loss choice no longer matters (0.791–0.794). Text with all banks is the largest single step.

Table S46: Head input (pooled feature, last-layer token mean, three depths, both) and head recipe (base: 40 epochs, lr 10^{-3}, wd 10^{-4}, batch 256; reg: 12 epochs, lr 3×10^{-4}, wd 10^{-2}, batch 64), single head or bag with projection, with text.

Study	Arm	bAcc	ECE	NLL	Recall G	False G
recipe	last reg bag nap+text	.792	.151	.575	.764	.100
recipe	pooled base bag nap+text	.791	.155	.581	.639	.070
input	pooled bag nap+text	.791	.155	.581	.639	.070
recipe	last base bag nap+text	.787	.150	.582	.678	.078
input	last bag nap+text	.787	.150	.582	.678	.078
recipe	pooled reg bag nap+text	.787	.155	.580	.783	.109
input	pooled last bag nap+text	.782	.154	.594	.707	.093
recipe	pooled reg single+text	.759	.136	.621	.688	.109
recipe	last reg single+text	.759	.133	.616	.650	.098
recipe	last base single+text	.757	.132	.616	.547	.074
input	last single+text	.757	.132	.616	.547	.074
recipe	pooled base single+text	.752	.143	.622	.531	.071
input	pooled single+text	.752	.143	.622	.531	.071
input	pooled last single+text	.744	.149	.631	.546	.085
input	depths bag nap+text	.731	.118	.642	.613	.118

Table S46: (continued)

Study	Arm	bAcc	ECE	NLL	Recall G	False G
input	depths single+text	.696	.139	.678	.520	.126

Table S47. Initial design on each held-out development site.

Held-out site	bAcc	Recall G	False G
Bangladesh Eye	.751	.716	.149
DeepDRiD	.776	–	.081
GRAPE	.812	.812	–
IDRiD	.758	–	.026
Jichi	.767	–	.101
OIA-DDR	.730	–	.202
SUSTech-SYSU	.946	–	.040

Table S48. Designs tried and not retained (development). Reference: pooled-feature pipeline 0.791; single cross-entropy head 0.675.

Experiment	bAcc	Reading
Batch network predicting temperature and class bias (1 seed)	.539	Below the raw head (.666)
EM prior shift with TS (1 seed)	.523	Unreliable with few classes per source
Polar Fourier features around the disc	.702	Below the pair loss alone (.723)
Site prior, κ from 30 to 3000	.693–.717	No gain
Full residual corrector over logits	.667	Recall G fell to 0
Corrector with fusion weights only	.781	ECE .057, bAcc did not rise
RPAttn head on three token depths (1 seed)	.664	Recall G .008; with text .690
RPAttn + MixStyle (1 seed)	.614	Recall G .012; with text .688
N/D from RPAttn, G from low-capacity branch (1 seed)	.788	.793 for the pooled pipeline on that seed
Late fusion of nine encoder inputs	.765	Other encoders lowered the score
Greedy nested encoder selection	.789	Mostly kept MedSigLIP alone
Magnified optic-disc view with whole image	.768	Recall G up (.767), false G up (.135)
Image-conditioned fusion gate	.763	Below a plain average
Global class offset chosen on other sites	.679	GRAPE fell to .006
ConvNeXt-T end-to-end with pair loss (1 seed)	.622	GRAPE .004

Reading. Heavier or learned modules repeatedly lost the glaucoma class at held-out sites, which led to the low-capacity design.

S8.2 Error analysis and disc-side bias

Reading. On OIA-DDR, 1,629 of 5,639 normal images were predicted G although the source holds no glaucoma. Between 0.9% and 13.7% of a site's images were wrong under all nine inputs; among normal IDRiD images, 28%. A global class offset did not fix the G confusion, so it is a ranking problem.

Table S49. Held-out development sites for the pooled-feature pipeline: head, text and fusion bAcc, recalls, and images wrong under all nine encoder inputs.

Held-out site	*n*	Head	Text	Fusion	N	D	G	All wrong
Bangladesh Eye	3,153	.651	.732	.753	.761	.963	.536	.039
OIA-DDR	11,269	.660	.738	.735	.685	.786	–	.137
DeepDRiD	1,804	.803	.721	.795	.682	.908	–	.111
GRAPE	528	.644	.750	.741	–	–	.741	.104
IDRiD	462	.757	.752	.751	.527	.974	–	.100
Jichi	8,475	.779	.736	.803	.824	.781	–	.103
SUSTech-SYSU	1,097	.975	.889	.957	.938	.975	–	.009

Table S50. Normal images predicted G, by disc side (automatic, label-free disc location), held-out site.

Held-out site	Disc left	Disc right
OIA-DDR	.386	.191
Bangladesh Eye	.141	.023
IDRiD	.119	.045
SUSTech-SYSU	.086	.019
Jichi	.067	.054
DeepDRiD	.096	.086

Table S51: Upgrades of the initial design's input on development data: dimension (384 or 1152), mirrored training and averaging, NAP k, weight decay. A separate study with its own runs; its "d384 plain" baseline is the initial design's configuration but differs slightly from the main development ablation (0.789 against 0.792).

Arm	bAcc	NLL	Recall G	False G	N→G left	N→G right
d384 mirror k32 wd0.1	.792	.577	.658	.078	.127	.127
d384 mirror k16 wd0.1	.792	.577	.658	.078	.127	.127
d384 mirror k32 wd0.01	.792	.577	.652	.076	.125	.124
d384 mirror k16 wd0.01	.792	.577	.652	.076	.125	.124
d384 mirror k32 wd0.001	.792	.578	.653	.077	.124	.124
d384 mirror k16 wd0.001	.792	.578	.653	.077	.124	.124
d384 mirror k8 wd0.001	.790	.578	.650	.077	.126	.125
d384 mirror k8 wd0.01	.790	.579	.648	.077	.126	.124
d384 mirror k8 wd0.1	.790	.578	.644	.077	.125	.124
d384 plain k32 wd0.01	.789	.576	.768	.103	.215	.141
d1152 mirror k32 wd0.01	.778	.603	.711	.100	.176	.165
d1152 plain k32 wd0.01	.770	.603	.823	.134	.296	.188

Reading. Mirroring removes the left-right gap and lowers false G but costs recall G; all 1152 dimensions are worse; k and weight decay hardly matter.

Table S52. Post-processing checked on held-out development sites, fitted on calibration rows of the other sites.

State	ECE	NLL	Cov 80 cov.	Cov 80 risk	Risk 10 risk	LAC cov.	LAC size	Class-LAC	APS cov./size
Raw	.148	.576	.720	.105	.094	.866	1.18	.846	.930/1.62
TS	.052	.502	.719	.104	.096	.867	1.18	.842	.946/1.78
FTS	.053	.505	.720	.106	.096	.867	1.18	.843	.919/1.61

S8.3 Patch-level evidence

The head input is the mean of patch tokens, and standardization, projection and the linear head are all linear, so each head's logits split exactly into per-patch contributions: $z_s(x) = \frac{1}{P}\sum_p W_s(I - U_s U_s^\top)x_p + b_s$ for the initial design, and the same without the projection for LOTUS. The maps below were computed with the initial design. Figure S5 maps the glaucoma evidence (G logit minus the larger of N and D, averaged over heads) for the most and least confident glaucoma image of each glaucoma test source. Tokens recomputed for these images matched the stored ones to within 0.13 (feature scale about 90).

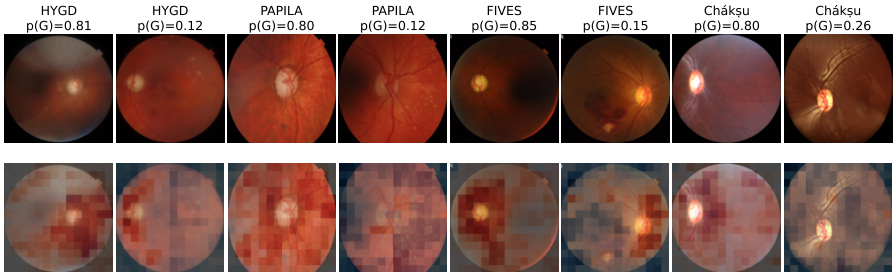


Fig. S5. Glaucoma evidence per patch (red: for G, blue: against), for the most and least confident glaucoma image of each source, $p(\mathrm{G})$ from raw fused logits. In confident cases the evidence concentrates on the optic disc; in missed cases it is weak and diffuse. Text scores are global and not mapped.

Reading. The maps are exact decompositions of the head, not post-hoc saliency, but they are not a validated localization of pathology.

S8.4 Failure cases

Reading. Errors are often confident (0.82–1.00 after TS), which a global temperature cannot catch. Three patterns recur. Glaucoma missed as normal or DR occurs on sources with strong colour casts or low contrast (HRF, AFIO), where the disc is less distinct. Normal images called glaucoma show a prominent, bright disc. Normal or DR confusions on Paraguay and PALM involve visible bright or dark spots in images labelled normal, which may reflect label scope (Paraguay graded only DR) rather than model error; without re-grading we cannot tell. These observations are descriptive, not a clinical review.

S8.5 Later attempts

Two further components were tried with selection rules written before their results were read; label scope was tested with the initial design, and the shift-aware reference is

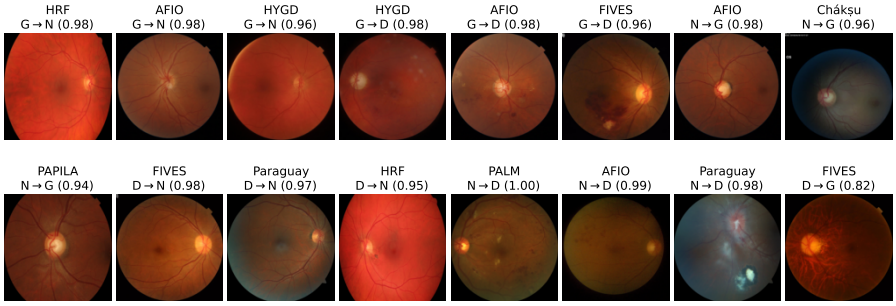


Fig. S6. Most confident errors of the initial design on test and Chákṣu (seed 0, after TS), up to three sources per error type: true→predicted class and confidence.

reported for LOTUS (Table S54; the initial design kept the risk target on 44 of 50 rotations against 23). *Label scope*: normal labels from DR-only sources were treated as the set {N, G}, with likelihood $p_{\mathrm{N}} + p_{\mathrm{G}}$. *Shift-aware reference*: temperature, thresholds and conformal quantiles were fitted on leave-one-source-out predictions rather than on Cal-B, with an optional normal certificate.

Table S53. Label scope, tested with the initial design: development (scope-aware bAcc counts a DR-only normal predicted G as correct) and the 50 rotations.

Arm	bAcc	Scope-aware bAcc	ECE	Recall G	N→G, DR-only	N→G, full scope
exact	.792	.844	.151	.764	.145	.322
scope	.776	.864	.136	.811	.244	.255
exact (rotations)	.771	.801	.033	.667	–	.248
scope (rotations)	.761	.831	.040	.797	–	.443

Table S54: LOTUS thresholds from Cal-B against the leave-one-source-out reference, 50 rotations, with and without the normal certificate; Held: rotations where test risk stayed at or below 10%. Here Cal-B is used whole; the main rotation table fits on one half of Cal-B and reads the other, so its LOTUS values differ in the third decimal.

Fitted on	Target	Test cov.	Test risk	Held	FR	G→N	Chákṣu cov.	Chákṣu risk
Cal-B	cov. .80	.724	.102	–	.069	.062	.383	.293
Cal-B	cov. .90	.852	.136	–	.099	.116	.657	.336
Cal-B	risk .10	.695	.105	19/50	.070	.084	.418	.277
Cal-B + cert.	cov. .80	.596	.096	–	.037	.024	.232	.437
Cal-B + cert.	cov. .90	.660	.129	–	.041	.033	.366	.546
Cal-B + cert.	risk .10	.575	.099	22/50	.037	.032	.243	.424
LOSO reference	cov. .80	.833	.131	–	.094	.108	.616	.329
LOSO reference	cov. .90	.914	.157	–	.117	.150	.812	.357
LOSO reference	risk .10	.592	.075	42/50	.046	.038	.192	.252
LOSO reference + cert.	cov. .80	.665	.121	–	.038	.028	.343	.514
LOSO reference + cert.	cov. .90	.705	.146	–	.038	.028	.437	.585
LOSO reference + cert.	risk .10	.556	.073	45/50	.036	.026	.153	.286

Reading. Label scope over-calls glaucoma at new sources. The shift-aware reference keeps coverage and risk promises far more often; risk at matched coverage is identical for both, so the arms differ only in where the threshold sits.

S9 Statistical Comparisons

Table S55: Paired bootstrap over leakage groups on the published split, 2,000 draws, Holm within each metric; differences are LOTUS minus baseline. Adjusted $p = 0.009$ is the floor set by 2,000 draws after Holm correction. Intervals are conditional on these seven test sources and on LOTUS's test-based selection; they do not include source-level variance (bAcc SD 0.039 across rotations).

Metric	Baseline	Baseline value	Diff.	95% CI	Adj. p
bAcc (raw)	16-shot probe, MedSigLIP	.748	+0.082	[+0.067, +0.097]	0.009
bAcc (raw)	Linear probe, MedSigLIP	.728	+0.102	[+0.088, +0.116]	0.009
bAcc (raw)	Zero-shot, MedSigLIP	.709	+0.122	[+0.107, +0.136]	0.009
bAcc (raw)	ERM, DINOv2 LoRA	.707	+0.123	[+0.107, +0.138]	0.009
bAcc (raw)	Group DRO, ConvNeXt-T	.707	+0.124	[+0.109, +0.138]	0.009
bAcc (raw)	Fine-tune (NLL rule), ConvNeXt-T	.677	+0.154	[+0.136, +0.170]	0.009
bAcc (raw)	CORAL, ConvNeXt-T	.676	+0.154	[+0.137, +0.170]	0.009
bAcc (raw)	SWAD, ConvNeXt-T	.672	+0.158	[+0.141, +0.175]	0.009
bAcc (raw)	Fine-tune, Swin-T	.655	+0.176	[+0.157, +0.193]	0.009
bAcc (raw)	WiSE-FT, SigLIP2-L	.634	+0.196	[+0.177, +0.214]	0.009
bAcc (raw)	MLP probe, DINOv2-B	.598	+0.232	[+0.213, +0.250]	0.009
bAcc (raw)	CoOp ctx-init, BiomedCLIP	.497	+0.334	[+0.315, +0.352]	0.009
bAcc (raw)	CoOp + LP, SigLIP-B	.496	+0.335	[+0.316, +0.353]	0.009
bAcc (raw)	Tip-Adapter-F, BiomedCLIP	.495	+0.336	[+0.315, +0.356]	0.009
bAcc (raw)	CoOp, SigLIP-B	.491	+0.339	[+0.320, +0.357]	0.009
bAcc (raw)	Tip-Adapter, BiomedCLIP	.478	+0.352	[+0.332, +0.373]	0.009
ECE (TS)	Zero-shot, MedSigLIP	.032	-0.002	[-0.017, +0.013]	0.799
ECE (TS)	16-shot probe, MedSigLIP	.044	-0.014	[-0.026, -0.000]	0.091
ECE (TS)	Linear probe, MedSigLIP	.079	-0.050	[-0.066, -0.028]	0.009
ECE (TS)	Group DRO, ConvNeXt-T	.091	-0.062	[-0.078, -0.043]	0.009
ECE (TS)	Tip-Adapter, BiomedCLIP	.091	-0.062	[-0.076, -0.044]	0.009
ECE (TS)	ERM, DINOv2 LoRA	.115	-0.086	[-0.103, -0.066]	0.009
ECE (TS)	Fine-tune (NLL rule), ConvNeXt-T	.133	-0.104	[-0.121, -0.081]	0.009
ECE (TS)	CORAL, ConvNeXt-T	.136	-0.107	[-0.124, -0.084]	0.009
ECE (TS)	SWAD, ConvNeXt-T	.139	-0.109	[-0.127, -0.087]	0.009
ECE (TS)	Tip-Adapter-F, BiomedCLIP	.147	-0.118	[-0.135, -0.100]	0.009
ECE (TS)	CoOp ctx-init, BiomedCLIP	.150	-0.121	[-0.140, -0.099]	0.009
ECE (TS)	WiSE-FT, SigLIP2-L	.168	-0.139	[-0.156, -0.117]	0.009
ECE (TS)	Fine-tune, Swin-T	.171	-0.142	[-0.158, -0.119]	0.009
ECE (TS)	MLP probe, DINOv2-B	.172	-0.143	[-0.160, -0.119]	0.009
ECE (TS)	CoOp, SigLIP-B	.175	-0.146	[-0.164, -0.124]	0.009
ECE (TS)	CoOp + LP, SigLIP-B	.176	-0.147	[-0.164, -0.124]	0.009
NLL (TS)	16-shot probe, MedSigLIP	.644	-0.187	[-0.212, -0.162]	0.009
NLL (TS)	Zero-shot, MedSigLIP	.690	-0.234	[-0.254, -0.214]	0.009
NLL (TS)	Linear probe, MedSigLIP	.694	-0.238	[-0.271, -0.202]	0.009
NLL (TS)	Group DRO, ConvNeXt-T	.926	-0.469	[-0.536, -0.405]	0.009
NLL (TS)	ERM, DINOv2 LoRA	.944	-0.487	[-0.552, -0.430]	0.009
NLL (TS)	CORAL, ConvNeXt-T	1.005	-0.549	[-0.613, -0.487]	0.009
NLL (TS)	Tip-Adapter, BiomedCLIP	1.020	-0.563	[-0.592, -0.533]	0.009
NLL (TS)	Fine-tune (NLL rule), ConvNeXt-T	1.036	-0.579	[-0.650, -0.513]	0.009
NLL (TS)	SWAD, ConvNeXt-T	1.056	-0.599	[-0.672, -0.529]	0.009
NLL (TS)	Tip-Adapter-F, BiomedCLIP	1.099	-0.643	[-0.679, -0.608]	0.009
NLL (TS)	Fine-tune, Swin-T	1.174	-0.717	[-0.790, -0.645]	0.009
NLL (TS)	CoOp + LP, SigLIP-B	1.180	-0.723	[-0.777, -0.669]	0.009
NLL (TS)	CoOp, SigLIP-B	1.180	-0.724	[-0.776, -0.671]	0.009
NLL (TS)	CoOp ctx-init, BiomedCLIP	1.189	-0.733	[-0.789, -0.675]	0.009
NLL (TS)	WiSE-FT, SigLIP2-L	1.243	-0.787	[-0.863, -0.711]	0.009

Table S55: (continued)

Metric	Baseline	Baseline value	Diff.	95% CI	Adj. p
NLL (TS)	MLP probe, DINOv2-B	1.259	-0.802	[-0.897, -0.708]	0.009

Table S56. Configurations with development bAcc of at least 0.791, all evaluated on the published split; the one with the highest test bAcc (bold) became the initial design, and LOTUS is this configuration without the projection. All ten exceed the strongest validation-selected representative (0.748).

Configuration	Dev	Test bAcc	Eq-src	Recall G	NLL (TS)	Chákṣu
Mirror, weight decay 10^{-3}	.800	.809	.726	.696	.503	.714
Mirror, weight decay 10^{-2}	.799	.808	.727	.708	.502	.726
Mirror, weight decay 10^{-4}	.795	.807	.724	.681	.511	.703
Cross-site pair loss	.794	.809	.717	.722	.507	.694
Same-site pair loss	.792	.791	.696	.670	.524	.713
Tokens, weight decay 10^{-2} (selected)	.792	.817	.724	.765	.482	.731
Pooled feature, weight decay 10^{-4}	.791	.799	.706	.687	.520	.684
Pooled feature + tokens	.791	.812	.719	.719	.493	.701
Disc side canonicalised	.791	.805	.733	.679	.511	.637
Mirrored text only	.791	.800	.714	.679	.512	.704

S10 Reproducibility

Code, saved result tables and this document are archived at `https://doi.org/10.5281/zenodo.23979471` (GitHub: `https://github.com/dangaura-tech/fundusshift`). The README maps each result to its script and table file. LOTUS and its ablation over the rotations and regions are produced by `experiments/revision_runs.py` (saved in `results/revision`); its published-split evaluation and paired tests by `experiments/lotus_eval.py`, its rotation promises by `experiments/lotus_rotation.py`, and its per-source and disease-level tables by `experiments/review_tables.py` (saved in `results/lotus`). The evaluation protocol, including the selection rules written before each later attempt, is `docs/protocol.md`. Images are obtained from their original publishers; split tables are available from the author on request.